

DIFFERENTIATION



SYNOPSIS

Derivative:

- (i) A function $y = f(x)$ is said to be differentiable if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ exists finitely. This limit is}$$

usually denoted by $f'(x)$ or $\frac{dy}{dx}$.

- (ii) Let $y = f(x)$ be a function and if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists, then } f \text{ is said to be}$$

differentiable at 'a' and the limit is called the derivative of f at a . The derivative of ' f ' at a is denoted by any one of the forms

$$\left(\frac{dy}{dx} \right)_{x=a} \text{ (or) } f'(a)$$

Sum or difference rule:

$$\frac{d}{dx}(c_1 f(x) \pm c_2 g(x)) = c_1 \frac{d}{dx}(f(x)) \pm c_2 \frac{d}{dx}(g(x))$$

where c_1, c_2 are constants.

Product Rule:

- i) If $f(x), g(x)$ are two differentiable functions of ' x ' then

$$\frac{d}{dx}(f(x).g(x)) = f(x) \frac{d}{dx}(g(x)) + g(x) \frac{d}{dx}(f(x))$$

- ii) If $f(x), g(x)$ and $h(x)$ are three differentiable functions of ' x ' then

$$\frac{d}{dx}(f(x).g(x).h(x)) = g(x).h(x) \frac{d}{dx}(f(x)) +$$

$$f(x).h(x) \frac{d}{dx}(g(x)) + f(x).g(x) \frac{d}{dx}(h(x))$$

Quotient Rule:

- If $f(x), g(x)$ are two differentiable functions of ' x '

$$\text{then } \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{d}{dx}(f(x)) - f(x) \frac{d}{dx}(g(x))}{(g(x))^2}$$

$$\text{W.E-1: } \frac{d}{dx} \left[(x+1)(x^2+1)(x^4+1)(x^8+1) \right] =$$

$$(15x^p - 16x^q + 1)(x-1)^{-2} \Rightarrow (p, q) =$$

$$\text{Sol: } f(x) = (x+1)(x^2+1)(x^4+1)(x^8+1) = \frac{x^{16}-1}{x-1}$$

$$\Rightarrow f'(x) = \frac{16x^{15}(x-1) - (x^{16}-1)}{(x-1)^2}$$

$$p = 16, q = 15$$

Chain Rule:

- If ' y ' is a function of ' t ' and ' t ' is a function of

$$'x' \text{ i.e., } y=f(t) \text{ and } t=g(x) \text{ then } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

similarly, if $y=f(u)$, where $u=f(v)$, $v=h(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

Note: We can extend this rule to any number of functions.

$$\text{W.E-2: } \frac{d}{dx} \left(\sin^2(\log \sqrt{x}) \right) =$$

$$\text{Sol: } 2 \sin(\log \sqrt{x}) \cos(\log \sqrt{x}) \cdot \frac{1}{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}$$

Derivatives of Basic elementary functions:

$$\text{➤ i) } \frac{d}{dx}(\text{Constant}) = 0$$

$$\text{ii) } \frac{d}{dx}(x^n) = n.x^{n-1}$$

$$\text{iii) } \frac{d}{dx}(e^x) = e^x$$

$$\text{iv) } \frac{d}{dx}(a^x) = a^x \cdot \log a$$

$$\text{v) } \frac{d}{dx}(|x|) = \frac{|x|}{x}; \text{ if } x \neq 0$$

$$\text{vi) } \frac{d}{dx}(\log_e |x|) = \frac{1}{x}$$

$$\text{vii) } \frac{d}{dx}(\log_a |x|) = \frac{1}{x \log a}$$

$$\text{viii) } \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$\text{ix) } \frac{d}{dx}\left(\frac{1}{x}\right) = \frac{-1}{x^2}$$

$$\text{x) } \frac{d}{dx}\left(\frac{1}{x^n}\right) = \frac{-n}{x^{n+1}}$$

$$\text{xi) } \frac{d}{dx}([x]) = \begin{cases} 0, & \forall x \notin \mathbb{Z} \\ \text{does not exist,} & \forall x \in \mathbb{Z} \end{cases}$$

where $[\cdot]$ stands for greatest integer function.

W.E-3: For a real number 'y', Let $[y]$ denote the integral part of 'y'. Then derivative of the function

$$f(x) = \frac{\tan[x - \pi]\pi}{1 + [x]^2} \text{ is}$$

Sol: $[(x - \pi)]\pi$ is an integral multiple of π , hence

$$f(x) = 0 \Rightarrow f'(x) = 0$$

Derivatives of Trigonometric functions:

$$\text{➤ i) } \frac{d}{dx}(\sin x) = \cos x$$

$$\text{ii) } \frac{d}{dx}(\cos x) = -\sin x$$

$$\text{iii) } \frac{d}{dx}(\tan x) = \sec^2 x$$

$$\text{iv) } \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$\text{v) } \frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$\text{vi) } \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$$

Derivatives of Inverse Trigonometric functions :

$$\text{➤ i) } \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, x \in (-1, 1)$$

$$\text{ii) } \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, x \in (-1, 1)$$

$$\text{iii) } \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, x \in \mathbb{R}$$

$$\text{iv) } \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}, x \in \mathbb{R}$$

$$\text{v) } \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x| \cdot \sqrt{x^2 - 1}}, x < -1 \text{ or } x > 1$$

$$\text{vi) } \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x| \cdot \sqrt{x^2 - 1}}, x < -1 \text{ or } x > 1$$

W.E-4: If $y = \tan^{-1} \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{d(\cos^{-1} x)} =$

Sol: Let $x = \cos 2\theta \Rightarrow \theta = \frac{1}{2} \cos^{-1} x; (-1 < x < 1)$

$$y = \tan^{-1} \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$$

$$= \tan^{-1}(\tan \theta) = \theta = \frac{1}{2} \cos^{-1} x$$

$$\therefore y = \frac{1}{2} \cos^{-1} x \Rightarrow \frac{d(y)}{d(\cos^{-1} x)} = \frac{1}{2}$$

Derivatives of Hyperbolic functions:

$$\text{➤ i) } \frac{d}{dx}(\sinh x) = \cosh x$$

$$\text{ii) } \frac{d}{dx}(\cosh x) = \sinh x$$

$$\text{iii) } \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\text{iv) } \frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$$

$$\text{v) } \frac{d}{dx}(\sec hx) = -\sec hx \tanh x$$

$$\text{vi) } \frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \cdot \coth x$$

Derivatives of Inverse Hyperbolic functions:

$$\text{➤ i) } \frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\text{ii) } \frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}, \text{ for } x \notin (-1, 1)$$

$$\text{iii) } \frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}, \text{ for } x \in (-1, 1)$$

$$\text{iv) } \frac{d}{dx}(\text{Coth}^{-1}x) = \frac{1}{1-x^2}, \text{ for } x \in (-\infty, -1) \cup (1, \infty)$$

$$\text{v) } \frac{d}{dx}(\text{Sech}^{-1}x) = \frac{-1}{x\sqrt{1-x^2}}, \text{ for } x \in (0, 1)$$

$$\text{vi) } \frac{d}{dx}(\text{Cosec}^{-1}x) = \frac{-1}{|x|\sqrt{1-x^2}} \\ \text{for } x \in (-\infty, 0) \cup (0, \infty)$$

Derivative of a Determinant :

➤ If $y = \begin{vmatrix} u(x) & v(x) & w(x) \\ p(x) & q(x) & r(x) \\ \lambda(x) & \mu(x) & \gamma(x) \end{vmatrix}$ then $\frac{dy}{dx} =$

$$\begin{vmatrix} u'(x) & v'(x) & w'(x) \\ p(x) & q(x) & r(x) \\ \lambda(x) & \mu(x) & \gamma(x) \end{vmatrix} + \begin{vmatrix} u(x) & v(x) & w(x) \\ p'(x) & q'(x) & r'(x) \\ \lambda(x) & \mu(x) & \gamma(x) \end{vmatrix} + \begin{vmatrix} u(x) & v(x) & w(x) \\ p(x) & q(x) & r(x) \\ \lambda'(x) & \mu'(x) & \gamma'(x) \end{vmatrix} \text{ or similarly column wise}$$

W.E-5.

$$f(x) = \begin{vmatrix} 2\cos x & 1 & 0 \\ x - \frac{\pi}{2} & 2\cos x & 1 \\ 0 & 1 & 2\cos x \end{vmatrix} \Rightarrow f'(\pi) =$$

[EAM - 2010]

Sol: $f'(x) = \begin{vmatrix} -2\sin x & 0 & 0 \\ x - \frac{\pi}{2} & 2\cos x & 1 \\ 0 & 1 & 2\cos x \end{vmatrix} +$

$$\begin{vmatrix} 2\cos x & 1 & 0 \\ 1 & -2\sin x & 0 \\ 0 & 1 & 2\cos x \end{vmatrix} +$$

$$\begin{vmatrix} 2\cos x & 1 & 0 \\ x - \frac{\pi}{2} & 2\cos x & 1 \\ 0 & 0 & -2\sin x \end{vmatrix}$$

put $x = \pi$ we get $f'(\pi) = 2$

Parametric Differentiation:

➤ If $x=f(t)$ and $y=g(t)$ are the parametric equations of a curve then

$$\text{i) } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{g'(t)}{f'(t)}$$

$$\text{ii) } \frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{d^2y}{dx^2} \right) \left(\frac{dt}{dx} \right)$$

W.E-6: If $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ then $\frac{dy}{dx} =$

Sol: $\frac{dx}{d\theta} = -3a \cos^2 \theta \cdot \sin \theta$

and $\frac{dy}{d\theta} = 3a \sin^2 \theta \cdot \cos \theta$

then $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = -\tan \theta$

Derivative of Implicit Functions:

➤ If $f(x, y) = 0$, then differentiate each term w.r.t. x regarding y as a function of x and then

collect the terms of $\frac{dy}{dx}$ together on left hand side and remaining terms on

right handside and then find $\frac{dy}{dx}$.

Alternative method : If $f(x, y) = 0$

then $\frac{dy}{dx} = - \frac{\left(\frac{\partial f}{\partial x} \right)}{\left(\frac{\partial f}{\partial y} \right)}$

$$= - \frac{\text{Partial derivative of } f \text{ w.r.t. } x}{\text{Partial derivative of } f \text{ w.r.t. } y}$$

W.E-7: If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$,

then $\frac{dy}{dx} = \frac{-(ax + hy + g)}{hx + by + f}$

Logarithmic Differentiation:

➤ If $y = \{f_1(x)\}^{f_2(x)}$ or

$$y = f_1(x) \cdot f_2(x) \cdot f_3(x) \dots \text{ or}$$

$$y = \frac{f_1(x) \cdot f_2(x) \cdot f_3(x) \dots}{g_1(x) \cdot g_2(x) \cdot g_3(x) \dots}$$

then take logarithm on both sides and differentiate both sides w.r.t. x

W.E-8: $x^y = y^x$ then $\frac{dy}{dx} =$

Sol: Take logarithm, we get, $y \log x = x \log y$
differentiating w.r.t x, we get

$$\log x \frac{dy}{dx} + y \cdot \frac{1}{x} = 1 \cdot \log y + x \cdot \frac{1}{y} \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y}{x} \cdot \frac{(x \log y - y)}{(y \log x - x)}$$

Derivative of Composite Function:

➤ $y = (g \circ f)(x) \Rightarrow y = g(f(x))$

$$\Rightarrow \frac{dy}{dx} = g'(f(x)) \cdot f'(x)$$

W.E-9: If $f(x) = e^x$, $g(x) = \sin^{-1}x$ and

$$h(x) = f(g(x)), \text{ then } \frac{h'(x)}{h(x)} =$$

Sol: $h'(x) = f'(g(x)) \cdot g'(x)$

$$\Rightarrow \frac{h'(x)}{h(x)} = \frac{f'(g(x)) \cdot g'(x)}{f(g(x))} = \frac{1}{\sqrt{1-x^2}}$$

Standard Derivatives:

➤ i) If $y = \sqrt[n]{f(x)} + \sqrt[n]{f(x)} + \sqrt[n]{f(x)} + \dots \text{ to } \infty$ then

$$\frac{dy}{dx} = \frac{f'(x)}{ny^{n-1} - 1}$$

ii) If $y = \sqrt{f(x)} + \sqrt{f(x)} + \sqrt{f(x)} + \dots \text{ to } \infty$ then

$$\frac{dy}{dx} = \frac{f'(x)}{(2y-1)}$$

iii) Let $f(x)$, $g(x)$ be the two differentiable functions of 'x' and if $y = f(x)^{g(x)}$ then

$$\frac{dy}{dx} = y \left(g'(x) \log f(x) + g(x) \frac{f'(x)}{f(x)} \right)$$

iv) If $y = f(x) + \frac{1}{y}$ then $\frac{dy}{dx} = \frac{y^2 f'(x)}{y^2 + 1}$

v) If $f(x+y) = f(x) \cdot f(y) \forall x, y \in \mathbb{R}$ and $f(x) \neq 0$ then $f'(x) = f'(0) \cdot f(x)$.

W.E-10: If $f(x+y) = f(x) \cdot f(y) \forall x \in \mathbb{R}$,

$$f(5)=2, f'(0)=3 \text{ then } f'(5) =$$

Sol: $f'(5) = f(5) \cdot f'(0) = 2(3) = 6$.

vi) If $f(x) = |x|$, then $f'(0)$ does not exist.

vii) $\frac{d}{dx} \left(\frac{ax+b}{cx+d} \right) = \frac{ad-bc}{(cx+d)^2}$

viii) $\frac{d}{dx} \left(\frac{af(x)+b}{cf(x)+d} \right) = \frac{(ad-bc)f'(x)}{\{cf(x)+d\}^2}$

ix) $\frac{d}{dx} \{ \log_e f(x) \} = \frac{f'(x)}{f(x)}$

x) $\frac{d}{dx} \{ \sqrt{f(x)} \} = \frac{f'(x)}{2\sqrt{f(x)}}$

xi) $\frac{d}{dx} \left\{ \frac{1}{f(x)} \right\} = \frac{-f'(x)}{(f(x))^2}$

xii) $\frac{d}{dx} \{ (f(x))^n \} = n \{ f(x) \}^{n-1} \cdot f'(x)$

xiii) If $y = \tan^{-1} \left(\frac{f(x) \pm g(x)}{1 \mp f(x) \cdot g(x)} \right)$ then

$$\frac{dy}{dx} = \frac{f'(x)}{1 + \{f(x)\}^2} \pm \frac{g'(x)}{1 + \{g(x)\}^2}$$

xiv) If $y = f(x)$ and $z = g(x)$ then

$$\frac{dy}{dz} = \frac{f'(x)}{g'(x)}$$

xv) If $y = f(x)^y$ then $\frac{dy}{dx} = \frac{y^2 \cdot f'(x)}{f(x) \{ 1 - y \log_e f(x) \}}$

Substitutions:

➤ While differentiating the given function using trigonometric transformation, observe the following points.

i) If the function involve the term $\sqrt{a^2 - x^2}$,

then put $x = a \sin \theta$ (or) $x = a \cos \theta$

ii) If the function involve the term $\sqrt{a^2 + x^2}$,

then put $x = a \tan \theta$ (or) $x = a \cot \theta$

iii) If the function involve the term $\sqrt{x^2 - a^2}$,

then put $x = a \sec \theta$ (or) $x = a \operatorname{cosec} \theta$

iv) If the function involve the term

$$\sqrt{\frac{a-x}{a+x}} \text{ (or) } \sqrt{\frac{a+x}{a-x}}, \text{ then}$$

put $x = a \cos \theta$ (or) $x = a \cos 2\theta$

W.E-11: $y = \tan^{-1} \left(\frac{x}{\sqrt{1+x^2}-1} \right)$, then $\frac{dy}{dx} =$

Sol: Put $x = \tan \theta$ then

$$y = \tan^{-1} \left(\cot \frac{\theta}{2} \right)$$

$$= \frac{\pi}{2} - \frac{1}{2} \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2(1+x^2)}$$

Higher Order Derivatives of functions:

➤ i) $\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$

ii) $\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right)$

W.E-12: If $y = \sin(\log_e x)$, then

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = \quad \text{(EAM-2008)}$$

Sol: $\frac{dy}{dx} = \frac{\cos(\log x)}{x} \Rightarrow x \frac{dy}{dx} = \cos(\log_e x)$

$$\Rightarrow x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = \frac{-\sin(\log_e x)}{x}$$

$$\Rightarrow x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -y$$



LEVEL - I (C.W)

FORMULA BASED

1. If $y = (\log x)^3$ then $\frac{dy}{dx} =$

1) $\frac{3(\log x)^2}{x}$ 2) $3(\log x)^2$

3) $\frac{(\log x)^2}{x}$ 4) $(\log x)^2$

2. $\frac{d}{dx} \left(\sqrt{\sin \sqrt{x}} \right) =$

1) $\frac{\cos \sqrt{x}}{4\sqrt{x \sin \sqrt{x}}}$ 2) $\frac{\sin \sqrt{x}}{4\sqrt{x \cos \sqrt{x}}}$

3) $\frac{-\cos \sqrt{x}}{4\sqrt{x \sin \sqrt{x}}}$ 4) $\frac{-\tan \sqrt{x}}{4\sqrt{x \cos \sqrt{x}}}$

3. If $y = a^{a^x}$ then $\frac{dy}{dx} =$

1) $y \cdot a^x (\log a)^2$ 2) $y \cdot a^x \cdot \log a$
3) $(y \cdot a^x)^2$ 4) $(y \cdot a^x)$

4. $\frac{d}{dx} [\log \{ \log(\log x) \}] =$

1) $\frac{1}{x \log x \log(\log x)}$ 2) $\frac{-1}{x \log x \log(\log x)}$

3) $\frac{x}{\log x \log(\log x)}$ 4) $\frac{1}{\log x \log(\log x)}$

INVERSE TRIGONOMETRIC FUNCTIONS AND SUBSTITUTIONS

5. If $y = 2^{ax}$ and $\frac{dy}{dx} = \log 256$ at $x=1$, then the value of a is

1) 0 2) 1
3) 2 4) 3

6. If $y = \tan^{-1} \left(\cot \left(\frac{\pi}{2} - x \right) \right)$ then $\frac{dy}{dx} =$

1) 1 2) -1 3) 0 4) $\frac{1}{2}$

7. If $y = \tan^{-1} (\sec x + \tan x)$ then $\frac{dy}{dx} =$

1) 1 2) $\frac{1}{2}$ 3) -1 4) 0

8. If $y = \tan^{-1}\left(\frac{(3-x)\sqrt{x}}{1-3x}\right)$ then $\frac{dy}{dx} =$

- 1) $\frac{3}{(1+x)\sqrt{x}}$ 2) $\frac{3}{2(1+x)\sqrt{x}}$
3) $\frac{-3}{2(1+x)\sqrt{x}}$ 4) 0

LOGARITHMIC DIFFERENTIATION

9. If $y = (\log x)^x$ then $\frac{dy}{dx} =$

- 1) $(\log x)^x \left[\log(\log x) + \frac{1}{\log x} \right]$
2) $(\log x)^x \left[\log(\log x) - \frac{1}{\log x} \right]$
3) $-(\log x)^x \left[\log(\log x) + \frac{1}{\log x} \right]$
4) $-(\log x)^x \left[\log(\log x) - \frac{1}{\log x} \right]$

10. If $y = (\sin x)^x$ then $\frac{dy}{dx} =$

- 1) $y(\log(\sin x) + x \cot x)$
2) $y(\log(\sin x) - x \cot x)$
3) $-y(\log(\sin x) - x \cot x)$
4) $-y(\log(\sin x) + x \cot x)$

11. If $x^y = e^{x-y}$ then $\frac{dy}{dx} =$

- 1) $\frac{\log x}{(1 + \log x)^2}$ 2) $-\frac{\log x}{(1 + \log x)^2}$
3) $-\frac{\log x}{(1 - \log x)^2}$ 4) $\frac{\log x}{(1 - \log x)^2}$

DERIVATIVE OF HYPERBOLIC AND INVERSE FUNCTIONS

12. $\frac{d}{dx} [\log(\cosh x)] =$

- 1) $\sinh x$ 2) $\coth x$ 3) $\tanh x$ 4) $-\coth x$

13. $\frac{d}{dx} (\operatorname{cosech}(x^3)) =$

- 1) $-\operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (3x^2)$
2) $\operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (3x^2)$
3) $\operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (6x^2)$
4) $-\operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (6x^2)$

14. $\frac{d}{dx} (\sinh^{-1}(3x)) =$

- 1) $\frac{1}{\sqrt{1+9x^2}}$ 2) $\frac{2}{\sqrt{1+9x^2}}$
3) $\frac{1}{\sqrt{1+9x^2}}$ 4) $\frac{1}{\sqrt{1+9x^2}}$

PARAMETRIC DIFFERENTIATION

15. If $x = a(t + \sin t)$, $y = a(1 - \cos t)$ if $\frac{dx}{dy} = \cot p$ then

$p =$

- 1) t 2) $2t$ 3) $\frac{t}{2}$ 4) $-t^2$

16. If $x = a(\cos t + \log(\tan \frac{t}{2}))$, $y = a \sin t$ then $\frac{dy}{dx} =$

(EAM-2008)

- 1) $\sin t$ 2) $\cot t$ 3) $\tan t$ 4) $\tan^2 t$

17. If $x = a \cos^4 t$, $y = b \sin^4 t$ then $\frac{dy}{dx}$ at $t = \frac{3\pi}{4}$ is

- 1) $\frac{-b}{a}$ 2) $\frac{b}{a}$ 3) $\frac{a}{b}$ 4) $\frac{-a}{b}$

DERIVATIVE OF ONE FUNCTION

W.R.T. ANOTHER FUNCTION

18. The derivative of a^x w.r.t. $\sin^{-1} x$ is

- 1) $a^x \cdot \log_e a \sqrt{1-x^2}$ 2) $\frac{a^x \cdot \log_e a}{\sqrt{1-x^2}}$
3) $\frac{a^x}{\sqrt{1-x^2}}$ 4) $\frac{1}{\sqrt{1-x^2}}$

19. The derivative of $\cos^{-1} \frac{1-x^2}{1+x^2}$ w.r.t.

$\tan^{-1} \frac{2x}{1-x^2}$ is

- 1) 0 2) 1 3) 2 4) $\frac{1}{2}$

DIFFERENTIATION

20. Derivative of $\sin^{-1}x$ with respect to $\cos^{-1}\sqrt{1-x^2}$

- 1) $\frac{1}{\sqrt{1-x^2}}$ 2) $\cos^{-1}x$ 3) 1 4) 0

DERIVATIVE OF IMPLICIT FUNCTION

21. If $2x^2-3xy+y^2+x+2y-8=0$, then $\frac{dy}{dx}$ is equal to
(EAM-2007)

- 1) $\frac{3y-4x-1}{2y-3x+2}$ 2) $\frac{3y+4x+1}{2y+3x+2}$
3) $\frac{3y-4x+1}{2y-3x-2}$ 4) $\frac{3y-4x+1}{2y+3x+2}$

22. If $x+y=\sin(x+y)$ then $\frac{dy}{dx} =$

- 1) $\frac{1}{2}$ 2) 0 3) -1 4) $\frac{1}{3}$

23. If $\sin(x+y)=\log(x+y)$, then $\frac{dy}{dx} =$

- 1) 2 2) -2 3) 1 4) -1

LEVEL-I (C.W)-KEY

- 1) 1 2) 1 3) 1 4) 1 5) 3 6) 1
7) 2 8) 2 9) 1 10) 1 11) 1 12) 3
13) 1 14) 3 15) 3 16) 3 17) 1 18) 1
19) 2 20) 3 21) 1 22) 3 23) 4

LEVEL-I (C.W)-HINTS

- $3(\log x)^2 \cdot \frac{1}{x}$
- $\frac{1}{2\sqrt{\sin \sqrt{x}}} \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$
- $a^x \log a \cdot \frac{d}{dx}(x)$
- $\frac{1}{\log(\log x)} \cdot \frac{1}{\log x} \cdot \frac{1}{x}$
- Take logarithm and differentiate
- $y = \tan^{-1}(\tan x) = x$
- $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$
- Put $\sqrt{x} = \tan \theta$

JEE-MAIN-JR-MATHS VOL-IV

- Take logarithm on both sides & differentiate.
- Take logarithm on both sides & differentiate.
- Taking logarithm on both sides & differentiate

12. $\frac{1}{\cosh x} \cdot \sinh x$

13. $-\operatorname{cosech}(x^3) \cdot \cot h(x^3) \cdot (3x^2)$

14. $\frac{d}{dx}(\sinh^{-1}x) = \frac{1}{\sqrt{1+x^2}}$

15. $\frac{dx}{dy} = \frac{\left(\frac{dx}{dt}\right)}{\left(\frac{dy}{dt}\right)} = \cot p$

16. $\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$

17. $\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$ and substitute $t = \frac{3\pi}{4}$

18. Let $y = a^x, z = \sin^{-1}x, \frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$

19. Put $x = \tan \theta$

20. $y = \sin^{-1}x, z = \cos^{-1}\sqrt{1-x^2} (x = \sin \theta)$

21. $\frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$ 22. $\frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$

23. $\cos(x+y) \left(1 + \frac{dy}{dx}\right) = \frac{1}{x+y} \left(1 + \frac{dy}{dx}\right)$



LEVEL - I (H.W)

FORMULA BASED

1. $\frac{d}{dx} \left\{ e^{\log \sqrt{1+\cot^2 x}} \right\} =$

- 1) $\operatorname{cosec} x \cot x$ 2) $-\operatorname{cosec} x \cdot \cot x$
3) $\operatorname{cosec}^2 x \cdot \cot x$ 4) 0

2. $\frac{d}{dx}(\sin \sqrt{2x-3}) =$
 1) $\frac{\cos \sqrt{2x-3}}{\sqrt{2x-3}}$ 2) $\frac{-\cos \sqrt{2x-3}}{2\sqrt{2x-3}}$
 3) $\sqrt{2x-3} \cos \sqrt{2x-3}$ 4) $\cos \sqrt{2x-3}$
3. If $f(x) = \frac{a^x}{x^a}$ then $f'(a) =$
 1) $\log a - 1$ 2) $\log a - a$
 3) $a \log a - a$ 4) $a \log a + a$
4. $\frac{d}{dx} \{x^n \cdot \log x\} =$
 1) $x^n(1+n \log x)$ 2) $x^{n-1}(1+n \log x)$
 3) $x^{n-1}(1-n \log x)$ 4) $x^{n-1}(1-n \log x)$

INVERSE TRIGONOMETRIC FUNCTIONS AND SUBSTITUTIONS

5. If $y = \tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$ then $\frac{dy}{dx} =$
 1) 1 2) -1 3) $\frac{-1}{2}$ 4) $\frac{1}{3}$
6. $\frac{d}{dx} \left\{ \tan^{-1} \frac{x}{1+x^2} + \tan^{-1} \frac{1+x^2}{x} \right\} =$
 1) 0 2) 1 3) $\frac{1}{2}$ 4) 2
7. If $y = \tan^{-1}\left(\frac{3a^2x - x^3}{a^3 - 3ax^2}\right)$ then $\frac{dy}{dx} =$
 1) $\frac{3a}{a^2+x^2}$ 2) $\frac{1}{a^2+x^2}$
 3) $\frac{-3a^2}{a^2+x^2}$ 4) $\frac{-3a}{a^2+x^2}$

LOGARITHMIC DIFFERENTIATION

8. If $y = 2^{2^x}$ then $\frac{dy}{dx} =$
 1) $y(\log 2)^2 2^x$ 2) $y(\log 2) \cdot 2^x$
 3) $y 2(\log 2)^2 2^x$ 4) $-y(\log 2) \cdot 2^x$
9. If $x^m \cdot y^n = (x+y)^{m+n}$ then $\frac{dy}{dx} =$
 1) $\frac{y}{x}$ 2) $-\frac{y}{x}$ 3) $\frac{my}{x}$ 4) $\frac{ny}{x}$

10. If $y = x^{\cos x}$ then $\frac{dy}{dx} =$
 1) $x^{\cos x} \left(\frac{\cos x}{x} + \sin x \cdot \log x \right)$
 2) $x^{\cos x} \left(\frac{\cos x}{x} - \sin x \cdot \log x \right)$
 3) $-x^{\cos x} \left(\frac{\cos x}{x} - \sin x \cdot \log x \right)$
 4) $-x^{\cos x} \left(\frac{\cos x}{x} + \sin x \cdot \log x \right)$

DERIVATIVE OF HYPERBOLIC AND INVERSE FUNCTIONS

11. $\frac{d}{dx} \left(\cosh^{-1} \frac{x}{2} \right) =$
 1) $\frac{1}{\sqrt{x^2+4}}$ 2) $\frac{1}{\sqrt{x^2-4}}$ 3) $\frac{-1}{\sqrt{x^2+4}}$ 4) $\frac{-1}{\sqrt{x^2-4}}$
12. $\frac{d}{dx} \left(\sinh^{-1}(\sqrt{x}) \right) =$
 1) $\frac{1}{\sqrt{1+x}} \cdot \frac{1}{\sqrt{x}}$ 2) $\frac{1}{\sqrt{1+x}} \cdot \frac{1}{2\sqrt{x}}$
 3) $-\frac{1}{\sqrt{1+x}} \cdot \frac{1}{2\sqrt{x}}$ 4) $-\frac{1}{\sqrt{1+x}} \cdot \frac{1}{\sqrt{x}}$
13. $\frac{d}{dx} (\sec h(\log x)), (x > 0)$
 1) $-\sec h(\log x) \cdot \tanh(\log x) \cdot \frac{1}{x}$
 2) $\sec h(\log x) \cdot \tanh(\log x) \cdot \frac{1}{x}$
 3) $\sec h(\log x) \cdot \coth(\log x) \cdot \frac{1}{x}$
 4) $-\sec h(\log x) \cdot \coth(\log x) \cdot \frac{1}{x}$

PARAMETRIC DIFFERENTIATION

14. If $x = at^2, y = 2at$ then $\frac{dy}{dx} =$
 1) $1/t$ 2) t 3) t^2 4) 1

15. If $x=a(1-\cos \theta)$, $y=a(\theta+\sin \theta)$ then $\frac{dy}{dx}$ at

$\theta=\frac{\pi}{2}$ is

- 1) 1 2) $\frac{1}{2}$ 3) 0 4) $-\frac{1}{2}$

16. If $x=a\cos^4 t$, $y=b\sin^4 t$ then $\frac{dy}{dx}$ at $t=\frac{\pi}{4}$ is

- 1) $-\frac{b}{a}$ 2) $\frac{b}{a}$ 3) $\frac{a}{b}$ 4) $-\frac{a}{b}$

DERIVATIVE OF ONE FUNCTION W.R.T. ANOTHER FUNCTION

17. The derivative of $e^{\sin^{-1} x}$ w.r.t. $\log x$ is

- 1) $\frac{-xe^{\sin^{-1} x}}{\sqrt{1-x^2}}$ 2) $\frac{xe^{\sin^{-1} x}}{\sqrt{1-x^2}}$
3) $\frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$ 4) $e^{\sin^{-1} x} \cdot \sqrt{1-x^2}$

18. The derivative of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ w.r.t. $\tan^{-1} x$ is

- 1) 0 2) 1 3) 2 4) $\frac{1}{2}$

19. The derivative of $\cos^{-1}(2x^2-1)$ with respect to $\cos^{-1} x$ is

- 1) 2 2) $\frac{2}{x}$ 3) $\frac{1}{2\sqrt{1-x^2}}$ 4) $\sqrt{1-x^2}$

DERIVATIVE OF IMPLICIT FUNCTION

20. If $ax^2 + 2hxy + by^2 = 0$ then $\frac{dy}{dx} =$

- 1) $-\left(\frac{ax+hy}{hx+by}\right)$ 2) $\left(\frac{ax+hy}{hx+by}\right)$
3) $-(ax+hy)(hx+by)$ 4) $(ax+hy)(hx+by)$

21. If $x^2 - y^2 = a(x-y)$ and $x \neq y$, then $\frac{dy}{dx} =$

- 1) 1 2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) -1

22. If $e^{x+y} = xy$ then $\frac{dy}{dx} =$

- 1) $\frac{y(1-x)}{x(y-1)}$ 2) $\frac{-y(1-x)}{x(y-1)}$ 3) $\frac{x(y-1)}{y(1+x)}$ 4) $\frac{-x(y-1)}{y(1+x)}$

LEVEL-I (H.W.)-KEY

- 1) 2 2) 1 3) 1 4) 2 5) 3 6) 1
7) 1 8) 1 9) 1 10) 2 11) 2 12) 2
13) 1 14) 1 15) 1 16) 1 17) 2 18) 4
19) 1 20) 1 21) 4 22) 1

LEVEL-I (H.W.)-HINTS

- $\frac{d}{dx}(e^{\log_e \csc x}) = \frac{d}{dx}(\csc x)$
- $\cos(\sqrt{2x-3}) \cdot \frac{1}{2\sqrt{2x-3}}$
- $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$
- $(uv)' = uv' + vu'$
- $y = \tan^{-1} \left(\frac{\sin(\frac{\pi}{2} - x)}{1 + \cos(\frac{\pi}{2} - x)} \right)$
- $\frac{d}{dx}(\tan^{-1} x + \cot^{-1} x) = \frac{d}{dx}(\frac{\pi}{2}) = 0$
- Put $x = a \tan \theta$
- $\frac{d}{dx}(a^x) = a^x \log a \cdot \frac{d}{dx}(x)$
- Take logarithm both sides and differentiate
- Take logarithm both sides and differentiate
- $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$
- $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$
- $\frac{d}{dx}(\sec hx) = -(\sec hx) \cdot (\tan hx)$
- $\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$
- $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)}$ and substitute $\theta = \frac{\pi}{2}$
- $\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$

17. $y = e^{\sin^{-1}x}$, $z = \log x$ and $\frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$

18. $\frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$, put $x = \tan \theta$

19. $y = \cos^{-1}(2x^2 - 1)$, $z = \cos^{-1}x$ and $\frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$

20. $\frac{dy}{dx} = \frac{-\left(\frac{\partial f}{\partial x}\right)}{\left(\frac{\partial f}{\partial y}\right)}$

21. $(x+y)(x-y) = a(x-y) \Rightarrow x+y = a$

22. $e^x e^y = xy$ and $\frac{dy}{dx} = \frac{-\left(\frac{\partial f}{\partial x}\right)}{\left(\frac{\partial f}{\partial y}\right)}$



LEVEL - II (C.W)

FORMULA BASED

1. $\frac{d}{dx}(\sin^5 x \cdot \sin 5x) =$

- 1) $\sin^4 x \cdot \sin 5x$ 2) $5\sin^4 x \cdot \sin 6x$
3) $5\sin^4 x \cdot \sin 5x$ 4) $-5\sin^4 x \cdot \sin 6x$

2. If $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$ then $\left(\frac{dy}{dx}\right)_{x=0} =$

- 1) $\frac{1}{2}$ 2) 1 3) -2 4) 2

3. If $y = e^{\sin^2 x + \sin^4 x + \sin^6 x + \dots + \infty}$, then $\frac{dy}{dx} =$

- 1) $e^{\tan^2 x}$ 2) $e^{\tan^2 x} \sec^2 x$
3) $2e^{\tan^2 x} \tan x \cdot \sec^2 x$ 4) 1

4. If $y = x + e^x$, then $\frac{d^2x}{dy^2}$ is equal to

- 1) $\frac{1}{(1+e^x)^2}$ 2) $\frac{-e^x}{(1+e^x)^2}$
3) $\frac{-e^x}{(1+e^x)^3}$ 4) e^x

5. If $x \cdot e^{xy} = y + \sin^2 x$, then at $x = 0$, $\frac{dy}{dx} =$

- 1) 1 2) 2 3) 3 4) 0

INVERSE TRIGONOMETRIC FUNCTIONS AND SUBSTITUTIONS

6. If $y = \tan^{-1}\left(\frac{2^x}{1+2^{2x+1}}\right)$ then $\frac{dy}{dx}$ at $x = 0$ is

- 1) $\frac{1}{10} \log 2$ 2) $\frac{1}{5} \log 2$ 3) $-\frac{1}{10} \log 2$ 4) $\log 2$

7. If $y = \sec(\tan^{-1}x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to

- 1) $\frac{1}{\sqrt{2}}$ 2) $\frac{1}{2}$ 3) 1 4) $\sqrt{2}$

8. $\frac{d}{dx}\left(\tan^{-1}\frac{x}{\sqrt{a^2-x^2}}\right) =$

- 1) $\frac{1}{\sqrt{a^2+x^2}}$ 2) $\frac{-1}{\sqrt{a^2+x^2}}$

- 3) $\frac{x}{\sqrt{a^2-x^2}}$ 4) $\frac{1}{\sqrt{a^2-x^2}}$

9. $\frac{d}{dx}\left\{\cot^{-1}\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right\} =$

- 1) $\frac{1}{\sqrt{1-x^2}}$ 2) $\frac{-1}{2\sqrt{1-x^2}}$ 3) $\frac{1}{1+x^2}$ 4) $\frac{-1}{2(1+x^2)}$

10. $\frac{d}{dx}\left(\sin^{-1}\left\{\frac{\sqrt{1+x}+\sqrt{1-x}}{2}\right\}\right) =$

- 1) $\frac{-1}{2\sqrt{1-x^2}}$ 2) $\frac{1}{2\sqrt{1-x^2}}$ 3) $\frac{1}{\sqrt{1-x^2}}$ 4) $\frac{-1}{\sqrt{1-x^2}}$

11. If $y = \cos^{-1}\left(\frac{x-x^{-1}}{x+x^{-1}}\right)$ then $\frac{dy}{dx} =$

- 1) $\frac{-2}{1+x^2}$ 2) $\frac{2}{1+x^2}$ 3) $\frac{1}{1+x^2}$ 4) $\frac{-1}{1+x^2}$

DERIVATIVE OF HYPERBOLIC AND INVERSE FUNCTIONS

12. $\frac{d}{dx} \left(\frac{3^x - 3^{-x}}{3^x + 3^{-x}} \right) =$

- 1) $\frac{4 \log 3}{(3^x + 3^{-x})^2}$ 2) $\frac{-4 \log 3}{(3^x + 3^{-x})^2}$
3) $\frac{1}{(3^x + 3^{-x})^2}$ 4) $\frac{-1}{(3^x + 3^{-x})^2}$

13. $\frac{d}{dx} (\tan^{-1}(\sinh x)) =$

- 1) cosechx 2) sechx 3) sinh x 4) cosh x

LOGARITHMIC DIFFERENTIATION

14. $\frac{d}{dx} \{x^{\tan x}\} =$

- 1) $x^{\tan x} \left\{ \frac{\tan x}{x} + \sec^2 x \cdot \log x \right\}$
2) $x^{\tan x} \left\{ \frac{\tan x}{x} - \sec^2 x \cdot \log x \right\}$
3) $x^{\tan x} \left\{ \frac{x}{\tan x} + \sec^2 x \cdot \log x \right\}$
4) $x^{\tan x} \left\{ \frac{x}{\tan x} - \sec^2 x \cdot \log x \right\}$

15. $\frac{d}{dx} [(\cos x)^{\log x} + (\log x)^x] =$

- 1) $(\log x)^x \left[\frac{1}{\log x} + \log(\log x) \right] + (\cos x)^{\log x} \left[\frac{1}{x} \log(\cos x) - \log x \cdot \tan x \right]$
2) $(\cos x)^{\log x} [\log(\cos x) - \cot x \cdot \log x] + (\log x)^{\log x} [1 + \log(\log x)]$
3) $((\cos x)^{\log x} + (\log x)^x) [\log x \cdot \cos x + x \log x]$
4) $(\log x)^x [\log x + \log(\log x)] (\cos x)^{\log x}$

16. The derivative of $y = (1-x)(2-x)(3-x) \dots (n-x)$ at $x = 1$ is

- 1) 0 2) $n! - 1$ 3) $(-1)^{n-1} (n-1)!$ 4) $(-1)(n-1)!$

PARAMETRIC DIFFERENTIATION

17. If $x = \sin^{-1} t$, $y = \log(1-t^2)$ then $\left(\frac{dy}{dx} \right)$ at $t = \frac{1}{2}$ is

- 1) $\frac{2}{\sqrt{3}}$ 2) $\frac{\sqrt{3}}{2}$ 3) $\frac{-2}{\sqrt{3}}$ 4) $\frac{-\sqrt{3}}{2}$

18. If $x = \frac{3at}{1+t^2}$, $y = \frac{3at^2}{1+t^2}$ then $\frac{dy}{dx} =$

- 1) $\frac{2t}{1-t^2}$ 2) $\frac{2t}{t^2-1}$ 3) $2t(t^2-1)$ 4) $-2t(t^2-1)$

19. If $x = \frac{1+t}{t^3}$ and $y = \frac{3+4t}{2t^2}$ then $x(y')^3 =$

- 1) $1-y'$ 2) $1+y'$ 3) $y'-1$ 4) y'

20. If $x = 2 \sin t$, $y = 2 \cos t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{2}$ is

- 1) 2 2) $-\frac{1}{2}$ 3) $-\frac{3}{4}$ 4) $-\frac{3}{2}$

21. If $x = \sin^{-1} t$ and $y = \log(1-t^2)$, then $\left. \frac{d^2y}{dx^2} \right|_{t=\frac{1}{2}}$ is

- 1) $-\frac{8}{3}$ 2) $\frac{8}{3}$ 3) $\frac{3}{4}$ 4) $-\frac{3}{4}$

DERIVATIVE OF ONE FUNCTION W.R.T. ANOTHER FUNCTION

22. The derivative of $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ w.r.t

$\tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$ at $x = 0$ is

- 1) $1/4$ 2) $1/8$ 3) $1/2$ 4) 1

23. Derivative of $e^{\sin^{-1} x}$ w.r.t. $\log x$ is

- 1) $-\frac{x e^{\sin^{-1} x}}{\sqrt{1-x^2}}$ 2) $\frac{x e^{\sin^{-1} x}}{\sqrt{1-x^2}}$ 3) $\frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$ 4) $e^{\sin^{-1} x} \cdot \sqrt{1-x^2}$

DERIVATIVE OF IMPLICIT FUNCTION

24. If $\sqrt{y - \sqrt{y - \sqrt{y - \dots}}} = \sqrt{x + \sqrt{x + \sqrt{x - \dots}}}$ then $\frac{dy}{dx}$ is equal to

- 1) $\frac{y-x+1}{y-x-1}$ 2) $\frac{y-x}{y+x}$ 3) $\frac{x+y+1}{y-x}$ 4) $\frac{y-x+1}{y+x}$

25. If $\cos y = x \cdot \cos(a+y)$ then $\frac{dy}{dx} =$

- 1) $\frac{\cos^2(a+y)}{\sin a}$ 2) $\frac{\cos^2(a+y)}{\cos a}$
3) $\frac{\cos a}{\sin^2(a+y)}$ 4) $\frac{\cos(a+y)}{\sin a}$

26. If $y = \sqrt{a^{3x+1}} + \sqrt{a^{3x+1}} + \sqrt{a^{3x+1}} + \dots \text{to } \infty$ then

$$\frac{dy}{dx} =$$

- 1) $\frac{a^{3x+1} \cdot \log a}{(2y-1)}$ 2) $\frac{3 \cdot a^{3x+1} \cdot \log a}{(2y-1)}$
3) $\frac{3 \cdot a^{3x+1} \cdot \log a}{(2y+1)}$ 4) $\frac{a^{3x+1} \cdot \log a}{(2y+1)}$

27. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ and $x \neq y$ then $\frac{dy}{dx} =$

- 1) $\frac{x+1}{(1-x)^2}$ 2) $\frac{1}{(1-x)^2}$
3) $\frac{-1}{(1+x)^2}$ 4) $\frac{1}{(1+x)^2}$

DERIVATIVE OF COMPOSITE FUNCTION

28. If $f(x) = \frac{x}{\sqrt{1-x^2}}$, $g(x) = \frac{x}{\sqrt{1+x^2}}$ then

$$\frac{d}{dx}(f \circ g(x)) =$$

- 1) 1 2) 0 3) -1 4) 2

LEVEL-II (C.W)-KEY

- 1) 2 2) 3 3) 3 4) 3 5) 1 6) 3
7) 1 8) 4 9) 2 10) 1 11) 1 12) 1
13) 2 14) 1 15) 1 16) 4 17) 3 18) 1
19) 2 20) 4 21) 1 22) 1 23) 2 24) 1
25) 1 26) 2 27) 3 28) 1

LEVEL-II (C.W)-HINTS

1. $\frac{d}{dx}(\sin^n x \cdot \sin nx) = n \sin^{n-1} x (\sin(n+1)x)$

2. $y = \sqrt{\frac{1 - \cos(\frac{\pi}{2} - 2x)}{1 + \cos(\frac{\pi}{2} - 2x)}}$

3. $y = e^{\sin^2 x (1 + \sin^2 x + \sin^4 x + \dots + \infty)} = e^{\sin^2 x (1 - \sin^2 x)^{-1}}$
 $= e^{\left(\frac{\sin^2 x}{\cos^2 x}\right)} = e^{\tan^2 x}$

4. $\Rightarrow \frac{dy}{dx} = 1 + e^x; \Rightarrow \frac{dx}{dy} = \frac{1}{1 + e^x}$

$$\Rightarrow \frac{d^2x}{dy^2} = \frac{d}{dy}\left(\frac{1}{1 + e^x}\right) \Rightarrow \frac{d^2x}{dy^2} = \frac{-1}{(1 + e^x)^2} \frac{d}{dy}(1 + e^x)$$

$$\Rightarrow \frac{d^2x}{dy^2} = \frac{-1}{(1 + e^x)^2} e^x \frac{dx}{dy} = \frac{-e^x}{(1 + e^x)^3}$$

5. If $x = 0$ then $y = 0$

$$e^{xy} + x \cdot e^{xy} \left[x \cdot \frac{dy}{dx} + y \right] = \frac{dy}{dx} + 2 \sin x \cos x$$

Put $x = 0, y = 0$, we get $\left(\frac{dy}{dx}\right)_{x=0} = 1$

6. $y = \tan^{-1}\left(\frac{2^{x+1} - 2^x}{1 + 2^x \cdot 2^{x+1}}\right)$

7. Given, $y = \sec(\tan^{-1}x)$

Let $\tan^{-1}x = \theta, \Rightarrow x = \tan \theta$

$\therefore y = \sec \theta = \sqrt{1 + x^2}$, differentiate w.r.t. x .

8. $x = a \sin \theta$

9. Put $x = \cos \theta$

10. Put $x = \cos \theta$

11. Put $x = \tan \theta$

12. $\frac{d}{dx}\left(\frac{e^{\log_e 3^x} - e^{\log_e 3^{-x}}}{e^{\log_e 3^x} + e^{\log_e 3^{-x}}}\right)$

13. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2}$

14. $\frac{d}{dx}\{f(x)^{g(x)}\} =$

$$f(x)^{g(x)} \left\{ g(x) \cdot \frac{1}{f(x)} \cdot f'(x) + \log f(x) \cdot g'(x) \right\}$$

15. $\frac{d}{dx}\{f(x)^{g(x)}\} =$

$$f(x)^{g(x)} \left\{ g(x) \cdot \frac{1}{f(x)} \cdot f'(x) + \log f(x) \cdot g'(x) \right\}$$

16. $\frac{dy}{dx} = (-1)(2-x)(3-x) \dots (n-x) +$

$$(-1)(1-x)(3-x) \dots (n-x) + \dots$$

$$(-1)(1-x)(2-x)\dots(n-x)$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{x=1} = (-1)(1)(2)\dots(n-1) = (-1)(n-1)!$$

$$17. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)} \text{ and substitute } t = \frac{1}{2}$$

$$18. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)}$$

$$19. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)}$$

$$20. \text{ Use } \frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dt} \times \frac{1}{\left(\frac{dx}{dt}\right)} \text{ and substitute } t = \pi/2$$

$$21. \text{ Use } \frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dt} \times \frac{1}{\left(\frac{dx}{dt}\right)} \text{ and substitute } t = 1/2$$

$$22. \text{ Put } x = \tan \theta \text{ in } \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

$$\text{Put } x = \sin \theta$$

$$\text{in } \tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$$

$$23. y = e^{\sin^{-1}x}; z = \log x; \frac{dy}{dz} = \frac{\left(\frac{dy}{dx} \right)}{\left(\frac{dz}{dx} \right)}$$

$$24. \sqrt{y-\sqrt{y-\sqrt{y-\dots\infty}}} = \sqrt{x+\sqrt{x+\dots\infty}} = v$$

$$\sqrt{y-v} = \sqrt{x+v} = v$$

$$\Rightarrow \left. \begin{matrix} y = v^2 + v \\ x = v^2 - v \end{matrix} \right\} \Rightarrow \frac{y+x}{2} = v^2 \text{ and } \frac{y-x}{2} = v$$

By eliminating v we get the solution.

$$25. x = \frac{\cos y}{\cos(a+y)} \text{ and then differentiate}$$

$$26. \frac{dy}{dx} = \frac{f'(x)}{(2y-1)}$$

$$27. x\sqrt{1+y} = -y\sqrt{1+x} \Rightarrow x^2(1+y) = y^2(1+x)$$

$$\Rightarrow x^2 - y^2 = xy^2 - x^2y$$

$$\Rightarrow (x-y)(x+y+xy) = 0, \text{ if } x \neq y$$

$$\Rightarrow x+y+xy = 0$$

$$\Rightarrow y(x+1) = -x \Rightarrow y = \frac{-x}{x+1}, \text{ differentiate}$$

$$28. (f \circ g)(x) = f \left(\frac{x}{\sqrt{1+x^2}} \right)$$



LEVEL - II (H.W)

FORMULA BASED

$$1. \frac{d}{dx} \left\{ \log(x + \sqrt{a^2 + x^2}) \right\} =$$

$$1) \frac{1}{(x + \sqrt{a^2 + x^2})} \quad 2) \frac{x}{\sqrt{a^2 + x^2}}$$

$$3) \frac{1}{x(x + \sqrt{a^2 + x^2})} \quad 4) \frac{1}{\sqrt{a^2 + x^2}}$$

$$2. \text{ If } y = \sqrt{\frac{\sec x - 1}{\sec x + 1}} \text{ then } \frac{dy}{dx} =$$

$$1) \frac{1}{2} \sec^2 \frac{x}{2} \quad 2) \sec^2 \frac{x}{2} \quad 3) \frac{1}{2} \tan \frac{x}{2} \quad 4) \tan \frac{x}{2}$$

$$3. \text{ If } y = 5^{2\{\log_5(x+1) - \log_5(3x+1)\}} \text{ then } \frac{dy}{dx} \text{ at } x=0 \text{ is}$$

$$1) 0 \quad 2) \frac{1}{3} \quad 3) \frac{3}{5} \quad 4) -4$$

$$4. \text{ If } y^2 = ax^2 + bx + c, \text{ then } y^3 \frac{d^2y}{dx^2} \text{ is}$$

$$1) \text{ a constant} \quad 2) \text{ a function of } x \text{ only} \\ 3) \text{ a function of } y \text{ only} \quad 4) \text{ a function of } x \text{ and } y$$

$$5. \text{ If } e^{g(y)} - e^{-g(y)} = 2f(x) \text{ then } \frac{dy}{dx} =$$

$$1) \frac{f'(x)}{g'(y)} \sqrt{1+f(x)^2} \quad 2) \frac{f'(x)}{g'(y)} \frac{1}{\sqrt{1+(f(x))^2}}$$

$$3) \frac{f'(x)}{g'(y)} \frac{1}{\sqrt{1+(f(x))^2}} \quad 4) 2 \frac{f'(x)}{g'(y)} \sqrt{1+f(x)^2}$$

INVERSE TRIGONOMETRIC FUNCTIONS AND SUBSTITUTIONS

$$6. y = \sin^{-1} 2x + \tan^{-1} 3x + \cos^{-1} 2x + \cot^{-1} 3x$$

$$\text{then } \frac{dy}{dx}$$

$$1) 0 \quad 2) 1 \quad 3) \frac{2}{\sqrt{1-x^2}} + \frac{3}{\sqrt{1+9x^2}} \quad 4) 3$$

7. If $y = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ then $\frac{dy}{dx} =$
- 1) $\frac{1}{1+x^2}$ 2) $\frac{-1}{1+x^2}$ 3) $\frac{1}{2(1+x^2)}$ 4) $\frac{2}{1+x^2}$
8. $\frac{d}{dx} \left\{ \tan^{-1} \left(\frac{a \sin x + b \cos x}{a \cos x - b \sin x} \right) \right\} =$
- 1) 0 2) 1 3) -1 4) $\frac{1}{1+x^2}$
9. $\frac{d}{dx} \left[\sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right] =$
- 1) 1 2) -1 3) $\frac{1}{2}$ 4) $\frac{-1}{2}$
10. $\frac{d}{dx} \left\{ \sin^{-1} \left(\frac{5x+12\sqrt{1-x^2}}{13} \right) \right\} =$
- 1) $\frac{1}{\sqrt{1-x^2}}$ 2) $\frac{-1}{\sqrt{1+x^2}}$
- 3) $\frac{2}{\sqrt{1-x^2}}$ 4) 0
11. $\frac{d}{dx} \left\{ \cos^{-1} x + \sin^{-1} \sqrt{1-x^2} \right\} =$
- 1) 0 2) 1 3) $\frac{2}{\sqrt{1-x^2}}$ 4) $\frac{-2}{\sqrt{1-x^2}}$

DERIVATIVE OF HYPERBOLIC AND INVERSE FUNCTIONS

12. $\frac{d}{dx} \left[\frac{e^{\sin \sqrt{x}} + e^{-\sin \sqrt{x}}}{e^{\sin \sqrt{x}} - e^{-\sin \sqrt{x}}} \right] =$
- 1) $\frac{-\frac{2}{\sqrt{x}} \cos \sqrt{x}}{(e^{\sin \sqrt{x}} - e^{-\sin \sqrt{x}})^2}$ 2) $\frac{-\frac{2}{\sqrt{x}} \cos \sqrt{x}}{e^{\sin \sqrt{x}}}$
- 3) $\frac{-2 \cos \sqrt{x}}{e^{\sin \sqrt{x}}}$ 4) $\frac{2 \cos \sqrt{x}}{e^{\sin \sqrt{x}}}$
13. $\frac{d}{dx} \left[\tanh^{-1} \left(\frac{2x}{1+x^2} \right) \right] =$
- 1) $\frac{2}{1-x^2}$ 2) $\frac{2}{x^2-1}$
- 3) $\frac{2}{1+x^2}$ 4) $\frac{-2}{x^2+1}$

LOGARITHMIC DIFFERENTIATION

14. $\frac{d}{dx} \left\{ (x^x)^x \right\} =$
- 1) $(x^x)^x \{x(1+\log x)\}$ 2) $(x^x)^x \cdot x(1-2\log x)$
- 3) $(x^x)^x (1+2\log x)x$ 4) $(x^x)^x \cdot x^2(1-2\log x)$
15. If $y = (ax+b)^{(cx+d)}$ then $\frac{dy}{dx} =$
- 1) $y \left\{ \frac{a(cx+d)}{(ax+b)} + c \log(ax+b) \right\}$
- 2) $y \left\{ \frac{a(cx+d)}{(ax+b)} - c \log(ax+b) \right\}$
- 3) $y \left\{ \frac{(cx+d)}{(ax+b)} + \log(ax+b) \right\}$
- 4) $y \left\{ \frac{(cx+d)}{(ax+b)} - \log(ax+b) \right\}$
16. If $(\cos x)^y = (\sin y)^x$ then $\frac{dy}{dx} =$
- 1) $\frac{\log(\sin y) + y \tan x}{\log(\cos x) - x \cot y}$ 2) $\frac{\log(\sin y) - y \tan x}{\log(\cos x) + \cot y}$
- 3) $\frac{\log(\sin y)}{\log(\cos x)}$ 4) $\frac{\log(\cos x)}{\log(\sin y)}$

PARAMETRIC DIFFERENTIATION

17. If $x = \cos^{-1} \left(\frac{1}{\sqrt{1+t^2}} \right)$, $y = \sin^{-1} \left(\frac{t}{\sqrt{1+t^2}} \right)$
- Then $\frac{dy}{dx} =$
- 1) 0 2) 1 3) -1 4) 2
18. $x = a \sin 2\theta (1 + \cos 2\theta)$,
 $y = b \cos 2\theta (1 - \cos 2\theta)$, then $\frac{dy}{dx} =$
- 1) $\frac{a}{b} \tan \theta$ 2) $\frac{b}{a} \tan \theta$
- 3) $\frac{a}{b} \cot \theta$ 4) $\frac{b}{a} \cot \theta$

19. If $x^2 + y^2 = t + \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$ then

$$x^3 y \frac{dy}{dx} = \quad \text{[EAM-2012]}$$

- 1) 0 2) 1 3) -1 4) 2

20. If $x = \cos^3 \theta, y = \sin^3 \theta$ then $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} =$ 1)

- $\tan^2 \theta$ 2) $\sec^2 \theta$ 3) $\sec \theta$ 4) $|\sec \theta|$

DERIVATIVE OF ONE FUNCTION W.R.T. ANOTHER FUNCTION

21. The derivative of $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$ with respect

$$\text{to } \sqrt{1+3x} \text{ at } x = -\frac{1}{3}$$

- 1) 0 2) $\frac{1}{2}$ 3) $\frac{1}{4}$ 4) $\frac{1}{8}$

DERIVATIVE OF IMPLICIT FUNCTION

22. If $y = \sqrt{x + \sqrt{y + \sqrt{x + \sqrt{y + \dots \infty}}}}$ then $\frac{dy}{dx}$ is equal to

1) $\frac{y^2 + x}{2y^3 - 2xy - 1}$ 2) $\frac{y^2 - x}{2y^3 - 2xy - 1}$

3) $\frac{y^2 - x}{2y^3 - 2xy + 1}$ 4) $\frac{y^2}{2y^3 - 2xy + 1}$

23. If $x \sin y = 3 \sin y + 4 \cos y$, then $\frac{dy}{dx} =$

1) $\frac{-\sin^2 y}{4}$ 2) $\frac{\sin^2 y}{4}$ 3) $\frac{-\cos^2 y}{4}$ 4) $\frac{\cos^2 y}{4}$

24. If $y = \sqrt{\log(\cos 2x) + y}$ then $\frac{dy}{dx} =$

1) $\frac{2 \tan 2x}{(2y-1)}$ 2) $\frac{\tan 2x}{(2y-1)}$ 3) $\frac{-2 \tan 2x}{(2y-1)}$ 4) $\frac{1}{(2y-1)}$

25. If $y = \frac{1}{x}$ then $\frac{dy}{\sqrt{1+y^4}} + \frac{dx}{\sqrt{1+x^4}} =$

1) 0 2) 1 3) $\frac{x}{y}$ 4) $\frac{y}{x}$

DERIVATIVE OF COMPOSITE FUNCTION

26. If g is the inverse of f and $f'(x) = \frac{1}{2+x^n}$, then

$g'(x)$ is equal to

- 1) $2+x^n$ 2) $2+(f(x))^n$ 3) $2+(g(x))^n$ 4) 0

LEVEL-II (H.W.)-KEY

- 1) 4 2) 1 3) 4 4) 1 5) 2 6) 1
7) 3 8) 2 9) 4 10) 1 11) 4 12) 1
13) 1 14) 3 15) 1 16) 1 17) 2 18) 2
19) 3 20) 4 21) 1 22) 2 23) 1 24) 3
25) 1 26) 3

LEVEL-II (H.W.)-HINTS

1. $\frac{d}{dx} \left\{ \log \left(x + \sqrt{a^2 + x^2} \right) \right\} = \frac{d}{dx} \left(\sinh^{-1} \frac{x}{a} \right) = \frac{1}{\sqrt{x^2 + a^2}}$

2. $y = \sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} = \tan \frac{x}{2}$

3. $y = 5^{2 \log_5 \left(\frac{x+1}{3x+1} \right)} = 5^{\log_5 \left(\frac{x+1}{3x+1} \right)^2} = \left(\frac{x+1}{3x+1} \right)^2$

4. $y^2 = ax^2 + bx + c \Rightarrow 2y \frac{dy}{dx} = 2ax + b \Rightarrow 2 \left(\frac{dy}{dx} \right)^2 + 2y \frac{d^2y}{dx^2} = 2a$

$$\Rightarrow y \frac{d^2y}{dx^2} = a - \left(\frac{dy}{dx} \right)^2; \Rightarrow y \frac{d^2y}{dx^2} = a - \left(\frac{2ax + b}{2y} \right)^2$$

$$\Rightarrow y \frac{d^2y}{dx^2} = \frac{4ay^2 - (2ax + b)^2}{4y^2}$$

$$4y^3 \frac{d^2y}{dx^2} = 4ac - b^2 \text{ is a constant}$$

5. $e^{g(y)} g'(y) \frac{dy}{dx} + e^{-g(y)} g'(y) \frac{dy}{dx} = 2f'(x)$

$$\frac{dy}{dx} = \frac{f'(x)}{g'(y)} \cdot \frac{1}{\sqrt{1 + (f(x))^2}}$$

6. $y = (\sin^{-1} 2x + \cos^{-1} 2x) + (\tan^{-1} 3x + \cot^{-1} 3x)$

$$y = \frac{\pi}{2} + \frac{\pi}{2} = \pi \Rightarrow \frac{dy}{dx} = 0$$

7. Put $x = \tan \theta$

8. Put $a = \cos \alpha, b = \sin \alpha$, then

$$\tan^{-1} \left(\frac{\sin(x + \alpha)}{\cos(x + \alpha)} \right) = \tan^{-1} (\tan(x + \alpha))$$

9. Put $x = \cos \theta$

10. $\frac{d}{dx} \left\{ \sin^{-1} \left(\frac{5}{13}x + \frac{12}{13}\sqrt{1-x^2} \right) \right\},$
 $x = \sin \theta, \cos \alpha = \frac{5}{13}, \sin \alpha = \frac{12}{13}$
11. Put $x = \cos \theta$
12. Given $= \frac{d}{dx} \left\{ \coth \left(\sin \sqrt{x} \right) \right\}$
13. $\frac{d}{dx} \left(\tanh^{-1} x \right) = \frac{1}{1-x^2}$
14. $y = (x^x)^x \Rightarrow \log y = x \log(x^x) = x^2 \log x$
 then differentiate w.r.t.x
15. $\frac{d}{dx} \left\{ f(x)^{g(x)} \right\} =$
 $f(x)^{g(x)} \left\{ g(x) \cdot \frac{1}{f(x)} \cdot f'(x) + \log f(x) \cdot g'(x) \right\}$
16. Take logarithm on both sides
17. Put $t = \tan \theta$
18. $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta} \right)}{\left(\frac{dx}{d\theta} \right)}$
19. $(x^2 + y^2)^2 = \left(t + \frac{1}{t} \right)^2$
 $\Rightarrow x^4 + y^4 + 2x^2y^2 = t^2 + \frac{1}{t^2} + 2$
 $\Rightarrow x^4 + y^4 + 2x^2y^2 = x^4 + y^4 + 2$
 $\Rightarrow x^2y^2 = 1$ and then differentiate
20. $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta} \right)}{\left(\frac{dx}{d\theta} \right)}$
21. $y = \sec^{-1} \left(\frac{1}{2x^2 - 1} \right), z = \sqrt{1 + 3x}$
 $(x = \cos \theta) \frac{dy}{dz} = \frac{\left(\frac{dy}{dx} \right)}{\left(\frac{dz}{dx} \right)}$
22. $y = \sqrt{x + \sqrt{y + y}} \Rightarrow y^2 = x + \sqrt{2y}$
23. $\sin y(x - 3) = 4 \cos y$ and then differentiate

24. Squaring on both sides and then differentiate.

25. $\frac{dy}{dx} = -\frac{1}{x^2}$

$$\frac{dy}{dx} + \frac{\sqrt{1+y^4}}{\sqrt{1+x^4}} = \frac{dy}{dx} + \frac{1}{x^2} = -\frac{1}{x^2} + \frac{1}{x^2} = 0$$

26. $g(x) = f^{-1}(x) \Rightarrow (f \circ g)(x) = x \Rightarrow f[g(x)]g'(x) = 1$

$$\Rightarrow f'[g(x)] = \frac{1}{g'(x)} \Rightarrow \frac{1}{2 + [g(x)]^n} = \frac{1}{g'(x)}$$

$$\text{then } g'(x) = 2 + [g(x)]^n$$

LEVEL - III

1. If $(f(x))^{g(y)} = e^{f(x)-g(y)}$ then $\frac{dy}{dx} =$

1) $\frac{f'(x) \log f(x)}{g(y)(1 + \log f(x))^2}$ 2) $\frac{f'(x) \log f(x)}{g^1(y)(1 + \log f(x))^2}$

3) $\frac{f(x) \cdot \log f(x)}{g^1(y)(1 - \log f(x))^2}$ 4) $2 \frac{f'(x) \log f(x)}{g(y)(1 + \log f(x))^2}$

2. If $\cos^{-1} \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = \log a$ then $\frac{dy}{dx} =$

1) $-\frac{x}{y}$ 2) $-\frac{y}{x}$ 3) $\frac{y}{x}$ 4) $\frac{x}{y}$

3. If $f(x) = (x - a)g(x)$ and $g(x)$ is continuous at $x = a$ then $f'(1) =$

1) $a g(1)$ 2) $g^1(1)$ 3) $g(1)$ 4) $a g^1(1)$

4. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ such that for all 'x' and y in $\mathbf{R}, |f(x) - f(y)| \leq |x - y|^3$ then $f(x)$

1) e^x 2) e^{-x} 3) x 4) 'c'(constant)

5. $\phi(x) = f(x)g(x)$ and $f'(x)g'(x) = k,$

then $\frac{2k}{f(x)g(x)} =$

1) $\frac{\phi''(x)}{\phi(x)} - \frac{f''(x)}{f(x)} - \frac{g''(x)}{g(x)}$ 2) $\frac{\phi''(x)}{\phi(x)} + \frac{f''(x)}{f(x)} + \frac{g''(x)}{g(x)}$

3) $\frac{\phi''(x)}{\phi(x)} + \frac{f''(x)}{f(x)} - \frac{g''(x)}{g(x)}$ 4) $\frac{\phi''(x)}{\phi(x)} - \frac{f''(x)}{f(x)} + \frac{g''(x)}{g(x)}$

6. If $f(x) = \frac{g(x)+g(-x)}{2} + \frac{2}{[h(x)+h(-x)]^{-1}}$
where g and h are differentiable functions

then $f'(0) =$

- 1) 1 2) 0 3) $\frac{1}{2}$ 4) $\frac{3}{2}$

7. If $x = \sqrt{\frac{1-t^2}{1+t^2}}$, $y = \frac{\sqrt{1+t^2} - \sqrt{1-t^2}}{\sqrt{1+t^2} + \sqrt{1-t^2}}$ then

$$\frac{dy}{dx} =$$

- 1) $\frac{1-x}{1+x}$ 2) $\frac{-2}{(1+x)^2}$ 3) $\frac{2}{(1+x)^2}$ 4) $\frac{1+x}{1-x}$

8. $\frac{d}{dx} \left\{ e^{\frac{1}{2} \log(1-\tanh^2 x)} + 3^{\frac{1}{2} \log_3(\cot^2 h^2 x - 1)} \right\} =$

- 1) $\frac{-(\sin h^3 x + \cos h^3 x)}{\cos h^2 x \sin h^2 x}$ 2) $\frac{\sin h^3 x + \cos h^3 x}{\cos h^2 x \sin h^2 x}$

- 3) $\frac{\sin h^2 x + \cos h^2 x}{\sin h^2 x \cos h^2 x}$ 4) $\sec h^2 x \cos h^2 x$

9. If $x < 1$, then

$$\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \dots \infty =$$

- 1) x 2) $\frac{1}{1+x}$ 3) $\frac{1}{1-x}$ 4) $\frac{1}{x}$

10. $f(x) = [Tan \pi / 4 + Tan x][Tan \pi / 4 + Tan(\pi / 4 - x)]$ and $g(x) = x(x+1)$. then the value of $g'[f(x)]$ is equal to

- 1) $\frac{1}{1+x^2}$ 2) 4 3) 5 4) $\frac{2x}{1+x^2}$

11. If $\sqrt{\tan y} = e^{\cos 2x} \cdot \sin x$, then $\frac{dy}{dx} =$

- 1) $\sin 2y \cdot (\cot x - 2 \sin 2x)$
2) $\sin 2x (\cot y - \sin y)$
3) $\sin 2y \cdot \sin 2x$ 4) $\cos 2y \cdot \cos 2x$

12. If $x = \cos \theta$, $y = \sin 5\theta$ then

$$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} =$$

- 1) $-5y$ 2) $5y$ 3) $25y$ 4) $-25y$

13. If $P(x) = ax^3 + bx^2 + cx + d$ and $p(0) = 4$,

$$P'(0) = 3, P''(0) = 4, P'''(0) = 6$$
 then

arrange the values of a, b, c, d in the descending order of their values

- 1) a, b, c, d 2) a, b, d, c
3) d, c, b, a 4) b, a, c, d

14. Let $f(x) = x^n$, n being a positive integer. The value of n for which $f'(a+b) = f'(a) + f'(b)$, when $a, b > 0$ is

- 1) 1 2) 2 3) 3 4) 4

15. If $f(x) = |x-2|$, $g(x) = f(f(x))$, then for $x > 4$, $g'(x) =$

- 1) 0 2) 1 3) -1 4) 2

16. If $\frac{d}{dx} \left(\frac{1+x^2+x^4}{1+x+x^2} \right) = ax + b$, then $(a, b) =$

- 1) $(-1, 2)$ 2) $(-2, 1)$ 3) $(2, -1)$ 4) $(1, 2)$

17. If $y = \log \cos \left(\tan^{-1} \left(\frac{e^x - e^{-x}}{2} \right) \right)$, then $\frac{dy}{dx} =$

- 1) $-\tanh x$ 2) $\sinh x$ 3) $\cosh x$ 4) $\coth x$

18. $\frac{d}{dx} \left[\log_e \left\{ (e^x + 2) + \sqrt{e^{2x} + 4e^x + 5} \right\} \right] =$

- 1) $\frac{1}{\sqrt{e^{2x} + 4e^x + 5}}$ 2) $\frac{e^x}{\sqrt{e^{2x} + 4e^x + 5}}$

- 3) $\frac{e^x}{\sqrt{e^{2x} + 4e^x + 3}}$ 4) $\frac{-e^x}{\sqrt{e^{2x} + 4e^x + 3}}$

19. If $f(x) = (ax + b) \cos x + (cx + d) \sin x$ and $f'(x) = x \cos x$ is an identity in x then a, b, c, d are given by

- 1) $a = b = c = d$ 2) 0, 1, -1, 0
3) 1, 0, -1, 0 4) 0, 1, 1, 0

20. If $f(x) = \begin{vmatrix} \sin x & 1 & 0 \\ x - \frac{\pi}{2} & \sin x & 1 \\ 0 & 1 & \sin x \end{vmatrix}$ then $\frac{df}{dx}$ at $x = \frac{\pi}{2}$ is

- 1) 2 2) $\frac{\pi}{2}$ 3) 1 4) 8

21. If $y = \log \left\{ \left(\frac{1+x}{1-x} \right)^{\frac{1}{4}} \right\} - \frac{1}{2} \tan^{-1}(x)$, then $\frac{dy}{dx} =$

- 1) $\frac{x}{1-x^2}$ 2) $\frac{x^2}{1-x^4}$ 3) $\frac{x}{1+x^4}$ 4) $\frac{x}{1-x^4}$

JEE-MAIN-JR-MATHS VOL-IV

22. If $f^1(x)=g(x)$ and $g^1(x)=-f(x)$ for all x and $f(2)=4=f^1(2)$ then $f^2(24)+g^2(24)$ is
1) 32 2) 24 3) 64 4) 48
23. If $f(x)=(\cos x+i \sin x)(\cos 3x+i \sin 3x) \dots (\cos(2n-1)x+i \sin(2n-1)x)$, then $f^{11}(x)$ is
1) $n^2 f(x)$ 2) $-n^4 f(x)$ 3) $-n^2 f(x)$ 4) $n^4 f(x)$
24. If $\sqrt{x+y}+\sqrt{y-x}=c$, then $\frac{d^2 y}{dx^2} =$
1) $\frac{2}{c}$ 2) $\frac{-2}{c^2}$ 3) $\frac{2}{c^2}$ 4) $\frac{-2}{c}$
25. Let $y=e^{2x}$. Then $\left(\frac{d^2 y}{dx^2}\right)\left(\frac{d^2 x}{dy^2}\right) =$
1) 1 2) e^{-2x} 3) $2e^{-2x}$ 4) $-2e^{-2x}$
26. If $f(x)=\sqrt{x^2-2x+1}$, then
1) $f'(x)=1, \forall x$ 2) $f'(x)=1, \forall x \geq 1$
3) $f'(x)=-1, \forall x \leq 1$
4) $f'(x)=1$ if $x>1$ and $f'(x)=-1$ if $x<1$
27. If $(\sin x)(\cos y)=1/2$ then $\frac{d^2 y}{dx^2}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is equal to
1) -4 2) -2 3) -6 4) 0
28. If $f(x)=\operatorname{sgn}(\cos x)$, then $f^1\left(\frac{\pi}{2}\right)$ is
1) 0 2) -1 3) 1 4) 2
29. If $y^2=p(x)$, a polynomial of degree 3, then $2 \frac{d}{dx}\left(y^3 \frac{d^2 y}{dx^2}\right)$ is equal to
1) $p^{111}(x)+p^1(x)$ 2) $p^{11}(x)+p^{111}(x)$
3) $p(x)p^{111}(x)$ 4) constant
30. If $y=\cos^{-1} \frac{9-x^2}{9+x^2}$ then $y^1(-1)$ is equal to
1) -3/5 2) 3/5 3) 2/7 4) 3/8
31. Let a function $y=y(x)$ be defined parametrically by $x=2t-|t|, y=t^2+t|t|$. Then $y^1(x), x>0$
1) 0 2) $4x$ 3) $2x$ 4) x
32. If $t=\frac{2\sqrt{2}-(1+\sqrt{3})}{\sqrt{3}-1}$ and $f(x)=\frac{2x}{1-x^2}, g(x)=\frac{3x-x^3}{1-3x^2}$ then $\frac{d}{dt}\{f(g(t))\} =$
1) 1 2) 2 3) 0 4) -1

DIFFERENTIATION

33. If $f(x)=\cos x \cos 2x \cos 4x \cos 8x \cos 16x$ then $f^1\left(\frac{\pi}{4}\right)$ equals
1) $-\sqrt{2}$ 2) 0 3) $\frac{1}{\sqrt{2}}$ 4) $\operatorname{Cosec} \frac{\pi}{4}$
34. If $f(x)=|x-8|$ & $g(x)=f(f(x)) \forall x>16$ then $g^1(x)$ equals
1) 1 2) -1 3) 0 4) 2
35. If $y^{y^{\infty}}=\log_e(x+\log_e(x+\dots))$ then $\frac{dy}{dx}$ at $(x=e^{-2}, y=\sqrt{2})$ is
1) $\frac{\log(\frac{e}{2})}{2\sqrt{2}(e^2-1)}$ 2) $\frac{\log 2}{2\sqrt{2}(e^2-1)}$
3) $\frac{\sqrt{2}\log(\frac{e}{2})}{(e^2-1)}$ 4) $\frac{\log(\frac{e}{2})}{(e^2-1)}$
36. If $y=\frac{\sin^2 x}{1+\cot x}+\frac{\cos^2 x}{1+\tan x}$ then $y^1(x)$ is equal to
1) $\cos 2x$ 2) $-\cos 2x$ 3) $2\cos^2 x$ 4) $\cos^3 x$
37. Let f be a function defined for every x , such that $f^1=-f, f(0)=0, f'(0)=1$ then $f(x)$ is equal to
1) $\tan x$ 2) e^x-1 3) $\sin x$ 4) $2\sin x$
38. If $t=\sin^{-1} 2^s$ then $\frac{ds}{dt}$ is equal to
1) $\frac{\log 2}{\sqrt{1-t^2}}$ 2) $\frac{\sin t}{\log 2}$ 3) $\frac{\cot t}{\log 2}$ 4) $\frac{\tan t}{\log 2}$
39. If $y=\frac{x}{a+\frac{x}{b+\frac{x}{a+\frac{x}{b+\dots\infty}}}}$ then $\frac{dy}{dx} =$
1) $\frac{1}{a(2y+b)}$ 2) $\frac{b}{a(2y+b)}$
3) $\frac{1}{ab(2y+b)}$ 4) $\frac{ab}{(2y+b)}$

40. If $f(x) = |x|^{\sin x}$, then $f^1\left(-\frac{\pi}{4}\right)$ is equals

- 1) $\left(\frac{\pi}{4}\right)^{\frac{1}{\sqrt{2}}}\left(\frac{\sqrt{2}}{2}\ln\frac{4}{\pi} - \frac{2\sqrt{2}}{\pi}\right)$
- 2) $\left(\frac{\pi}{4}\right)^{\frac{1}{\sqrt{2}}}\left(\frac{\sqrt{2}}{2}\ln\frac{4}{\pi} + \frac{2\sqrt{2}}{\pi}\right)$
- 3) $\left(\frac{\pi}{4}\right)^{\frac{1}{\sqrt{2}}}\left(\frac{\sqrt{2}}{2}\ln\frac{\pi}{4} - \frac{2\sqrt{2}}{\pi}\right)$
- 4) $\left(\frac{\pi}{4}\right)^{\frac{1}{\sqrt{2}}}\left(\frac{\sqrt{2}}{2}\ln\frac{\pi}{4} + \frac{2\sqrt{2}}{\pi}\right)$

41. If $Y = \tan^{-1}\left(\frac{1}{1+x+x^2}\right) + \tan^{-1}\left(\frac{1}{x^2+3x+3}\right) + \tan^{-1}\left(\frac{1}{x^2+5x+7}\right) + \dots$ upto n terms

then $y^1(0) =$

- 1) $\frac{-1}{1+n^2}$ 2) $\frac{-n^2}{1+n^2}$ 3) $\frac{n}{1+n^2}$ 4) n

42. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ then $\frac{dy}{dx} =$

- 1) $\sqrt{\frac{1-x^2}{1-y^2}}$ 2) $\sqrt{(1-x^2)(1-y^2)}$
- 3) $\sqrt{\frac{1-y^2}{1-x^2}}$ 4) $\frac{1}{\sqrt{(1-x^2)(1-y^2)}}$

43. If $f(x) = 3e^{x^2}$, then $f^1(x) - 2xf(x) + \frac{1}{3}f(0) - f^1(0)$ is equal to

- 1) 0 2) 1 3) $\frac{7}{3}e^{x^2}$ 4) e^{x^2}

44. If $f^1(x) = \sin(\log x)$ and $y = f\left(\frac{2x+3}{3-2x}\right)$, then $\frac{dy}{dx}$ is

- 1) $\sin(\log x) \cdot \frac{1}{x \log x}$ 2) $\frac{12}{(3-2x)^2} \sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$
- 3) $\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$ 4) $\sin(\log x)$

45. If $f(x) = \log_{x^2}(\log x)$, then $f^1(x)$ at $x=e$ is

- 1) 0 2) 1 3) $\frac{1}{e}$ 4) $\frac{1}{2e}$

46. $y = \log^n x$ where $\log^n$ means $\log \cdot \log \cdot \log \dots$ (repeated n times)

then $x \cdot \log x \cdot \log^2 x \cdot \log^3 x \dots \log^{n-1} x \cdot \frac{dy}{dx} =$

- 1) $\log x$ 2) $\log^n x$ 3) $\frac{1}{\log x}$ 4) 1

47. $\frac{d^2x}{dy^2} =$ [AIEEE - 2011]

- 1) $\frac{1}{\left(\frac{dy}{dx}\right)^2}$ 2) $\frac{\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^2}$ 3) $\left(\frac{d^2y}{dx^2}\right)$ 4) $\frac{-\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^3}$

48. If $y = \left(x + \sqrt{x^2 - a^2}\right)^n$ then $(x^2 - a^2) \left(\frac{dy}{dx}\right)^2 =$

- 1) n^2y 2) $-n^2y$ 3) ny^2 4) n^2y^2

49. If $|x| < 1$, then

$$\frac{d}{dx} \left[1 + \frac{p}{q}x + \frac{P(p+q)}{2} \left(\frac{x}{q}\right)^2 + \frac{p(p+q)(p+2q)}{3} \left(\frac{x}{q}\right)^3 + \dots \right] =$$

- 1) $\frac{p}{q(1-x)^{\frac{p}{q}+1}}$ 2) $\frac{p}{q(1-x)^{\frac{p}{q}}}$
- 3) $(1-x)^{-pq-1}$ 4) $(1-x)^{pq+1}$

50. $\frac{d}{dx} \left[\frac{\tan x - \cot x}{\tan x + \cot x} \right] =$

- 1) $2 \sin 2x$ 2) $-2 \sin 2x$ 3) $2 \cos 2x$ 4) $-2 \cos 2x$

51. If $y = \cot^{-1}(1+x^2-x)$, then $\frac{dy}{dx} =$

- 1) $\frac{1}{1+x^2} + \frac{1}{1+(x-1)^2}$ 2) $\frac{1}{1+(x-1)^2} - \frac{1}{1+x^2}$
- 3) $\frac{1}{1+x^2} - \frac{1}{1+(x-1)^2}$ 4) $-\frac{1}{1+x^2} - \frac{1}{1+(x-1)^2}$

52. $\frac{d}{dx} \left(\sqrt{\sec^{-1}x^2} \right) =$

- 1) $\frac{1}{x\sqrt{\sec^{-1}x^2} \cdot \sqrt{x^4-1}}$ 2) $\frac{x}{\sqrt{\sec^{-1}x^2} \cdot \sqrt{x^4-1}}$
- 3) $\frac{-1}{x\sqrt{\sec^{-1}x^2} \cdot \sqrt{x^4-1}}$ 4) $\frac{-1}{\sqrt{\sec^{-1}x^2} \cdot \sqrt{x^4-1}}$

53. $\frac{d}{dx} \left\{ \frac{1}{2} \cot h \frac{x}{2} - \frac{1}{6} \cot h^3 \frac{x}{2} \right\} =$

1) $-\frac{1}{4} \operatorname{cosec} h \frac{4x}{2}$ 2) $-\frac{1}{4} \cot h^4 x / 2$

3) $\frac{1}{4} \operatorname{cosec} h^4 x / 2$ 4) $\frac{1}{4} \cot h^4 x / 2$

54. If $x = \sec \theta - \cos \theta$ and $y = \sec^n \theta - \cos^n \theta$ then

$\left(\frac{dy}{dx} \right)^2$ is equal to

1) $\frac{n^2(y^2 + 4)}{x^2 + 4}$ 2) $\frac{n^2(y^2 - 4)}{x^2}$

3) $n \frac{(y^2 - 4)}{x^2 - 4}$ 4) $\left(\frac{ny}{x} \right)^2 - 4$

55. If $y = \sin x \left[\frac{1}{\sin x \cdot \sin 2x} + \frac{1}{\sin 2x \cdot \sin 3x} + \dots \right.$

$\left. + \frac{1}{\sin nx \sin(n+1)x} \right]$ then $\frac{dy}{dx} =$

1) $\cot x - \cot(n+1)x$

2) $(n+1) \operatorname{cosec}^2(n+1)x - \operatorname{cosec}^2 x$

3) $\operatorname{cosec}^2 x - (n+1) \operatorname{cosec}^2(n+1)x$

4) $\cot x + \cot(n+1)x$

56. The value of $3 f(x) - 2 f\left(\frac{1}{x}\right) = x$

then $f'(2) =$

1) $\frac{2}{7}$ 2) $\frac{1}{2}$ 3) 2 4) $\frac{7}{2}$

57. If $x = \exp \left\{ \tan^{-1} \left(\frac{y - x^2}{x^2} \right) \right\}$ then $\frac{dy}{dx} =$

1) $2x[1 + \tan(\log x)] + x \cdot \sec^2(\log x)$

2) $x[1 + \tan(\log x)] + \sec^2(\log x)$

3) $2x[1 + \tan(\log x)] + x^2 \sec^2(\log x)$

4) $2x[1 + \tan(\log x)] + \sec^2(\log x)$

58. If $af(\tan x) + bf(\cot x) = x$ then $f'(\cot x) =$

1) $\frac{1}{a-b}$ 2) $\frac{\sin^2 x}{a+b}$ 3) $\frac{\sin^2 x}{a-b}$ 4) $\frac{\sin^2 x}{b-a}$

59. If $x = a \sec^2 \theta, y = a \tan^3 \theta$ then $\frac{d^3 y}{dx^3} =$

1) $\frac{-3}{8a^2} \cot^3 \theta$ 2) $\frac{3}{8a^2} \cot^3 \theta$

3) $3 \sec^2 \theta \tan \theta$ 4) $\frac{3}{4a} \cot \theta$

60. Let $f(x) = x^5 + 2x^3 + 3x + 4$ then the value of

$28 \frac{d}{dx} (f^{-1}(x))$ at $x = -2$ is

1) 1 2) 2 3) 1/14 4) -2

61. If $f(x) = \cos x \cos 2x \cos 2^2 x \cos 2^3 x \dots \cos 2^{n-1} x$

and $n > 1$, then $f'\left(\frac{\pi}{2}\right)$ is

1) 1 2) 0 3) -1 4) $(-1)^{n-1}$

62. If g is the inverse function of f and

$f'(x) = \sin x$ then $g'(x)$ is

1) $\operatorname{cosec}(g(x))$ 2) $\sin(g(x))$

3) $-\frac{1}{\sin(g(x))}$ 4) $\cos(g(x))$

63. If $f(1) = 3, f'(1) = -\frac{1}{3}$, then the derivative

of $\{x^{11} + f(x)\}^{-2}$ at $x = 1$, is

1) $-\frac{1}{2}$ 2) -1 3) 1 4) $f'(1)$

64. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and

$g(x) = f^{-1}(x)$, then the value of $g'(1)$ is

1) $\frac{1}{2}$ 2) 2 3) 1 4) $-\frac{1}{2}$

65. If $\sin y + e^{-x \cos y} = e$ then $\frac{dy}{dx}$ at $(1, \pi)$ is equal to

1) $\sin y$ 2) $-x \cos y$ 3) e 4) $\sin y - x \cos y$

66. If $f(x) = \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec x & \tan x & x \\ 1 & \tan x - \tan \theta & 0 \end{vmatrix}$ then $f'(\theta)$ is

1) 0 2) -1 3) independent of θ 4) none

67. If $u = f(x^2)$, $v = g(x^3)$, $f'(x) = \sin x$, $g'(x) = \cos$

x then find $\frac{du}{dv}$

- 1) 1 2) $\frac{2}{3}$ 3) $\frac{2 \sin x^2}{3x \cdot \cos x^3}$ 4) $\frac{2x^2}{3x^3}$

68. If $f(0) = 0$, $f'(0) = 2$, then the derivative of $y = f(f(f(f(x))))$ at $x = 0$ is

- 1) 2 2) 8 3) 16 4) 4

69. If $x = \phi(t)$, $y = \psi(t)$ then $\frac{d^2y}{dx^2}$ is

- 1) $\frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^2}$ 2) $\frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^3}$
3) $\frac{\phi^{11}}{\psi^{11}}$ 4) $\frac{\psi^{11}}{\phi^{11}}$

70. If $u = x^2 + y^2$, and $x = s + 3t$, $y = 2s - t$, then

$\frac{d^2u}{ds^2}$ is equal to

- 1) 12 2) 32 3) 36 4) 10

71. Let $y = t^{10} + 1$, and $x = t^8 + 1$, then $\frac{d^2y}{dx^2}$ is

- 1) $\frac{5}{2}t$ 2) $20t^8$ 3) $\frac{5}{16t^6}$ 4) $\frac{15}{16t^6}$

72. $\frac{d}{dx} \left[\cos^2 \left(\tan^{-1} \left(\sin \left(\cot^{-1} x \right) \right) \right) \right] =$

- 1) $\frac{2}{(x^2+2)^2}$ 2) $\frac{2x}{(x^2+2)^2}$ 3) $\frac{x^2+1}{x^2+2}$ 4) $\frac{-2x}{(x^2-1)^2}$

73. If $x \sin y = \sin(y+a)$ and

$\frac{dy}{dx} = \frac{A}{1+x^2-2x \cos a}$ then the value of A is

- 1) 2 2) $\cos a$ 3) $\sin a$ 4) -2

74. If $y = \tan^{-1} \left(\frac{1}{\cos^2 x + \cos x + 1} \right) +$

$\tan^{-1} \left(\frac{1}{\cos^2 x + 3 \cos x + 3} \right) + \dots$ upto n terms then $f'(0) =$

- 1) 0 2) 1 3) $\frac{\pi}{2}$ 4) π

75. $f(x) = \cot^{-1} \left(\frac{x^x - x^{-x}}{2} \right)$ then $f'(1) =$

- 1) $-\log 2$ 2) $\log 2$ 3) 1 4) -1

76. If $y = \left(1 + \frac{1}{x} \right)^x$ then $\frac{2\sqrt{y_2(2) + \frac{1}{8}}}{\log \frac{3}{2} - \frac{1}{3}}$ is

- 1) 1 2) 3 3) 0 4) $\frac{1}{3}$

77. If $y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \cos x \dots \infty}}}$ then $\frac{dy}{dx} =$

- 1) $\frac{(1+y) \cos x + y \sin x}{1+2y + \cos x - \sin x}$ 2) $\frac{(1+y) \sin x + y \cos x}{1+2y + \cos x - \sin x}$

- 3) $\frac{(1+y) \cos x - y \sin x}{1+2y - \cos x + \sin x}$ 4) $\frac{(1+y) \cos x + y \sin x}{1+2y - \cos x - \sin x}$

78. If $\sqrt{\frac{v}{\mu}} + \sqrt{\frac{\mu}{v}} = 6$, then $\frac{dv}{d\mu} =$

- 1) $\frac{17\mu - v}{\mu - 17v}$ 2) $\frac{\mu - 17v}{17\mu - v}$

- 3) $\frac{17\mu + v}{\mu - 17v}$ 4) $\frac{\mu + 17v}{17\mu - v}$

79. If $y = \tan^{-1} \left(\frac{4 \sin 2x}{\cos 2x - 6 \sin^2 x} \right)$ then $\frac{dy}{dx}$ at $x = 0$ is

- 1) 10 2) 12 3) 6 4) 8

80. $y = \tan^{-1} \left(\frac{\log \left(\frac{e}{x^2} \right)}{\log(ex^2)} \right) + \tan^{-1} \left(\frac{3+2 \log x}{1-6 \log x} \right)$

then $\frac{d^2y}{dx^2} =$

- 1) 2 2) 1 3) 0 4) -1

81. $y = \cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right]$ then $\frac{dy}{dx} =$

- 1) $\frac{1}{2}$ 2) $\frac{2}{3}$ 3) 3 4) 1

82. If $x^2 + y^2 = 2$ and $y^{11} + A \cdot y^{-3} = 0$ then $A =$

- 1) 0 2) 2 3) 1 4) 3

83. If $p(x)$ is a polynomial such that

$p(x^2+1) = \{p(x)\}^2 + 1$ and $p(0) = 0$ then $p'(0)$ is equal to

- 1) -1 2) 0 3) 1 4) 2

84. $\frac{d}{dx} \left\{ e^{x^e} + x^{e^x} + e^{x^x} \right\} =$
 1) $e^{x^e} + x^{e^x} + e^{x^x}$ 2) $x^2 \cdot e^{x^e} + e^x \cdot x^{e^x} + x^x \cdot e^{x^x}$
 3) $e \cdot e^{x^e} \cdot x^{e-1} + x^{e^x} \cdot e^x \left(\frac{1}{x} + \log x \right) + e^{x^x} \cdot x^x (1 + \log x)$
 4) $e \cdot e^{x^e} \cdot x^{e-1} + x^{e^x} \cdot e^x \left(\frac{1}{x} - \log x \right) + e^{x^x} \cdot x^x (1 - \log x)$
85. If $\sin \theta \sin(2\alpha + \theta) \sin(4\alpha + \theta) \dots$
 $\sin(2(n-1)\alpha + \theta) = \frac{\sin n\theta}{2^{n-1}}$ where $2n\alpha = \pi$
 then
 $\cot(\theta) + \cot(2\alpha + \theta) + \cot(4\alpha + \theta) + \dots$
 $+ \cot(2(n-1)\alpha + \theta) =$
 1) $-n \cot n\theta$ 2) $n \cot n\theta$
 3) $n \tan n\theta$ 4) $-n \tan n\theta$
86. The derivative of $y = (1-x)(2-x)(3-x) \dots (n-x)$ at $x = 1$ is
 1) $n!$ 2) $(n-1)!$ 3) $-(n-1)!$ 4) 0
87. If $y = \sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}}$ then $\frac{dy}{d\theta}$ at $\theta = \frac{3\pi}{4}$ is
 1) -2 2) 2 3) 1 4) -1
88. If $\sqrt{1-x^{2n}} + \sqrt{1-y^{2n}} = a(x^n - y^n)$, then
 $\sqrt{\frac{1-x^{2n}}{1-y^{2n}}} \frac{dy}{dx}$ is equal to
 1) $\frac{x^{n-1}}{y^{n-1}}$ 2) $\frac{y^{n-1}}{x^{n-1}}$ 3) $\frac{x}{y}$ 4) 1
89. If $y = |\cos x| + |\sin x|$, then $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is
 1) $\frac{1-\sqrt{3}}{2}$ 2) 0 3) $\frac{\sqrt{3}-1}{2}$ 4) $\frac{\sqrt{3}+1}{2}$
90. If $f^{-1}(x) = \sqrt{2x^2 - 1}$ and $y = f(x^2)$, then $\frac{dy}{dx}$ at $x = 1$ is
 1) 2 2) 1 3) -2 4) -1
91. The first derivative of the function $\cos^{-1} \left(\sin \sqrt{\frac{1+x}{2}} \right) + x^x$ with respect to x at $x=1$ is
 1) $\frac{3}{4}$ 2) 0 3) $\frac{1}{2}$ 4) $-\frac{1}{2}$

92. If $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$, then the value of $\frac{d}{d(\tan \theta)} f(\theta)$ is
 1) 1 2) -1 3) 0 4) $\sqrt{2}$
93. If $y = \cos^{-1}(\cos x)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$ is
 1) 1 2) -1 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{5\pi}{4}$
94. If $y = f\left(\frac{3x+4}{5x+6}\right)$ and $f^{-1}(x) = \tan x^2$ then $\frac{dy}{dx}$ is equal to
 1) $-2 \tan \left(\frac{3x+4}{5x+6} \right)^2 \times \frac{1}{(5x+6)^2}$ 2) $f\left(\frac{3 \tan x^2 + 3}{5 \tan x^2 + 6}\right) \tan x^2$
 3) $\tan x^2$ 4) $2 \tan \left(\frac{3x+4}{5x+6} \right)^2 \times \frac{1}{(5x+6)^2}$
95. If $x = \operatorname{cosec} \theta - \sin \theta$, $y = \operatorname{cosec}^n \theta - \sin^n \theta$ then
 $(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - n^2 y^2 =$
 1) n^2 2) $2n^2$ 3) $3n^2$ 4) $4n^2$
96. Differential coefficient of $\left(x^{\frac{l+m}{m-n}} \right)^{\frac{1}{n-l}} \cdot \left(x^{\frac{m+n}{n-l}} \right)^{\frac{1}{l-m}} \cdot \left(x^{\frac{n+l}{l-m}} \right)^{\frac{1}{m-n}}$ w.r.t. x , is
 1) 1 2) 0 3) -1 4) x^{lmn}
97. Let $u(x)$ and $v(x)$ are differentiable function such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)} \right)' = q$, then $\frac{p+q}{p-q} =$
 1) 1 2) 0 3) 7 4) -7
98. If $f(x) = |\cos x - \sin x|$, then $f^{-1}\left(\frac{\pi}{4}\right) =$
 1) $\sqrt{2}$ 2) $-\sqrt{2}$ 3) 0 4) does not exist
99. A function is represented parametrically by the equations $x = \frac{1+t}{t^3}$, $y = \frac{3}{2t^2} + \frac{2}{t}$ then $\frac{dy}{dx} - x \left(\frac{dy}{dx} \right)^2$ has the absolute value equal to
 1) -1 2) 1 3) 0 4) 2

100. If $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$, then $f'(x) =$
 1) 0 2) $6x$ 3) $12x^3$ 4) $6x^2$
101. Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $y'(1)$ equals
 [AIEEE-2009]
 1) -1 2) 1 3) $\log 2$ 4) $-\log 2$
102. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x) + 2)]^2$, then $g'(0) =$
 [AIEEE 2010]
 1) -4 2) 0 3) -2 4) 4
103. Let f be twice differentiable function such that $g'(x) = -f(x)$ and $f'(x) = g(x)$,
 $h(x) = (f(x))^2 + (g(x))^2$. If $h(5) = 11$, then $h(10)$ is
 1) 22 2) 11 3) 0 4) 8
104. Given that $f(x)$ is a differentiable function of x and that $f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2$ and that $f(2) = 5$. Then $f(3)$ is equal to
 1) 10 2) 24 3) 15 4) 19
105. Let $f(x+y) = f(x)f(y)$ and $f(x) = 1 + (\sin 2x)g(x)$ where $g(x)$ is continuous. Then $f'(x)$ equals
 1) $f(x)g(0)$ 2) $2f(x)g(0)$ 3) $2g(0)$ 4) $2f(0)$

LEVEL-III-KEY

- 1) 2 2) 3 3) 3 4) 4 5) 1 6) 2
 7) 2 8) 1 9) 3 10) 3 11) 1 12) 4
 13) 3 14) 2 15) 2 16) 3 17) 1 18) 2
 19) 4 20) 3 21) 2 22) 1 23) 2 24) 3
 25) 4 26) 4 27) 1 28) 1 29) 3 30) 1
 31) 2 32) 3 33) 4 34) 1 35) 1 36) 2
 37) 3 38) 3 39) 2 40) 1 41) 2 42) 3
 43) 2 44) 2 45) 4 46) 4 47) 4 48) 4
 49) 1 50) 1 51) 3 52) 1 53) 3 54) 1
 55) 2 56) 2 57) 1 58) 3 59) 1 60) 2
 61) 1 62) 1 63) 4 64) 2 65) 3 66) 2
 67) 3 68) 3 69) 2 70) 4 71) 3 72) 2
 73) 3 74) 1 75) 4 76) 2 77) 1 78) 2
 79) 4 80) 3 81) 1 82) 2 83) 3 84) 3
 85) 2 86) 3 87) 2 88) 1 89) 3 90) 1
 91) 3 92) 1 93) 2 94) 1 95) 4 96) 2
 97) 1 98) 4 99) 2 100) 4 101) 1 102) 1
 103) 2 104) 1 105) 2

LEVEL-III-HINTS

1. $g(y) \log f(x) = f(x) - g(y)$
 $\Rightarrow \frac{dy}{dx} = \frac{f'(x) \cdot \log f(x)}{g'(y)(1 + \log f(x))^2}$
2. $\left(\frac{x^2 - y^2}{x^2 + y^2} \right) = \cos(\log a)$
 Applying Componendo and dividendo
3. $f(x) = (x - a)g(x)$, $f(a) = 0$
 $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{(x - a)g(x) - 0}{x - a} = g(a)$
4. $\lim_{x \rightarrow y} \frac{|f(x) - f(y)|}{|x - y|} \leq |x - y|^2$
 $f'(x) = 0 \Rightarrow f(x)$ is constant
5. Differentiate $\phi(x) = f(x)g(x)$ two times and simplify
6. Since $f(x)$ is an even function $f'(0) = 0$
7. $\frac{1}{x} = \frac{\sqrt{1+t^2}}{\sqrt{1-t^2}} \Rightarrow \frac{1-x}{1+x} = \frac{\sqrt{1+t^2} - \sqrt{1-t^2}}{\sqrt{1+t^2} + \sqrt{1-t^2}}$
 $\Rightarrow \frac{1-x}{1+x} = y$
8. $\frac{d}{dx} \{ \sec hx + \cosh x \}$
 $= -\sec hx \tanh x - \cosh x \coth x$
 $= -\left[\frac{\sinh x}{\cosh^2 x} + \frac{\cosh x}{\sinh^2 x} \right] = -\left[\frac{\sinh^3 x + \cosh^3 x}{\cosh^2 x \sinh^2 x} \right]$
9. $(1+x)(1+x^2) \dots (1+x^n) = \frac{1-x^{2n}}{1-x}$ and taking logarithms and differentiate
10. $f(x) = (1 + \tan x) \left(1 + \frac{1 - \tan x}{1 + \tan x} \right) = 2$
11. Taking Logarithm on both sides, then
 $\frac{dy}{dx} = \frac{-(\partial f / \partial x)}{(\partial f / \partial y)}$
12. $\frac{dy}{dx} = \frac{(dy/d\theta)}{(dx/d\theta)}$

13. $P(0) = 4 \Rightarrow d = 4$, $P^1(x) = 3ax^2 + 2bx + c$
at $x = 0$, $P^1(0) = 3$ then $c = 3$
similarly to find a and b calculate $P^{11}(0)$ and $P^{111}(0)$
14. Verify the options
15. If $x > 4$ then $|x - 2| = x - 2$
16. $\frac{d}{dx}(x^2 - x + 1) = ax + b$, $2x - 1 = ax + b$
 $\Rightarrow a = 2, b = -1$
17. Express 'y' as $\log \cos \tan^{-1}(\sinh x) = \log(\sec hx)$
and simplify
18. Express $\log \left\{ (e^x + 2) + \sqrt{(e^x + 2)^2 + 1} \right\}$ as
 $\sinh^{-1}(e^x + 2)$ and simplify
19. $f(x) = (ax + b)\cos x + (cx + d)\sin x$
 $f'(x) = (ax + b)(-\sin x) + \cos x \cdot a +$
 $(cx + d)\cos x + c \sin x$
Compare on Both sides
20. Find its determinant and put $x = \frac{\pi}{2}$.
21. $\log(a^m) = m \log a$,
 $\frac{d}{dx}(\log x) = \frac{1}{x}$, $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{(1 + x^2)}$
22. $\frac{d}{dx}(f^2(x) + g^2(x)) = 2[f(x)f'(x) + g(x)g'(x)]$
 $= 2[f(x)g(x) - g(x)f(x)] = 0$
hence $f^2(x) + g^2(x)$ is constant. Thus
 $f^2(24) + g^2(24) = f^2(2) + g^2(2) = 16 + 16 = 32$
23. $f(x) = \cos x \cdot \cos 3x \dots \cos(2n-1)x$
 $= \cos[x + 3x + \dots + (2n-1)x] = \cos[(1+3+\dots+(2n-1))x]$
 $= \cos(n^2x) \Rightarrow f(x) = \cos n^2x + i \sin n^2x$
differentiate w.r.t 'x' on both sides
24. consider $(\sqrt{x+y})^2 - (\sqrt{y-x})^2$
 $= (\sqrt{x+y} - \sqrt{y-x})(\sqrt{x+y} + \sqrt{y-x})$
 $2x = (\sqrt{x+y} - \sqrt{y-x}) \cdot c; \Rightarrow \sqrt{x+y} - \sqrt{y-x} = \frac{2x}{c} \rightarrow (1)$
given $\sqrt{x+y} + \sqrt{y-x} = c \rightarrow (2)$
 $(1) + (2): 2\sqrt{x+y} = \frac{2x}{c} + c$

- $4(x+y) = \frac{4x^2}{c^2} + c^2 + 4x$, then $4y = \frac{4x^2}{c^2} + c^2$
differentiate w.r.t 'x' two times
25. $\frac{dy}{dx} = 2e^{2x}$ and $\frac{d^2y}{dx^2} = 4e^{2x}$. Also $x = \frac{1}{2} \log y$.
so $\frac{dx}{dy} = \frac{1}{2y}$ and $\frac{d^2x}{dy^2} = -\frac{1}{2y^2} = -\frac{1}{2}e^{-4x}$
Hence $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right) = -2e^{-2x}$
26. $f(x) = \sqrt{x^2 - 2x + 1} = |x-1| = \begin{cases} x-1 & \text{if } x > 1 \\ 0 & \text{if } x = 1 \\ -(x-1) & \text{if } x < 1 \end{cases}$
 $f'(x) = 1$ if $x > 1$ and $f'(x) = -1$ if $x < 1$
27. Differentiating, we have,
 $\cos x \cos y - \sin x \sin y \frac{dy}{dx} = 0$.
Putting $x = y = \frac{\pi}{4}$, we have $\frac{dy}{dx}\bigg|_{(\frac{\pi}{4}, \frac{\pi}{4})} = 1$.
differentiating again, we have
 $-\sin x \cos y - \cos x \sin y \frac{dy}{dx} - \cos x \sin y \frac{dy}{dx}$
 $-\sin x \sin y \left(\frac{dy}{dx}\right)^2 - \sin x \sin y \frac{d^2y}{dx^2} = 0$,
Putting $x = y = \frac{\pi}{4}$, we have $\frac{d^2y}{dx^2}\bigg|_{(\frac{\pi}{4}, \frac{\pi}{4})} = -4$
28. $f(x) = 0$, $x = \frac{\pi}{2}$; $= 1$, $x < \frac{\pi}{2}$; $= -1$, $x > \frac{\pi}{2}$
29. $y^2 = p(x) \Rightarrow 2yy_1 = p'(x)$ then $2yy_2 + 2y_1y_1 = p^{11}(x)$
 $\Rightarrow 2[y^3y_2 + y^2y_1^2] = y^2p^{11}(x)$
 $\Rightarrow 2\left[y^3y_2 + \frac{(p'(x))^2}{4}\right] = y^2p^{11}(x)$
 $\Rightarrow 2y^3y_2 = y^2p^{11}(x) - \frac{(p'(x))^2}{2}$
 $\Rightarrow y^3y_2 = \frac{1}{2}\left[y^2p^{11}(x) - \frac{(p'(x))^2}{2}\right]$
differentiate w.r.t 'x' on both sides
30. $\cos^{-1} \frac{9-x^2}{9+x^2} = \cos^{-1} \frac{1-(\frac{x}{3})^2}{1+(\frac{x}{3})^2} = 2 \tan^{-1}(\frac{x}{3})$
If $0 \leq x < \infty$ and is equal to $-2 \tan^{-1}(\frac{x}{3})$
If $-\infty < x \leq 0$. Hence $y'(x) = \begin{cases} \frac{2}{1+(\frac{x}{3})^2} \cdot \frac{1}{3} & \text{if } 0 < x < \infty \\ -\frac{2}{1+(\frac{x}{3})^2} \cdot \frac{1}{3} & \text{if } -\infty < x \leq 0 \end{cases}$

Hence $y'(-1) = -\frac{3}{5}$

31. $x = 2t - |t| = \begin{cases} t, & t \geq 0 \\ 3t, & t < 0 \end{cases}$ so $t = \begin{cases} x, & x \geq 0 \\ \frac{x}{3}, & x < 0 \end{cases}$

therefore, we can express $y = t^2 + t|t| = \begin{cases} 2t^2, & x \geq 0 \\ 0, & x < 0 \end{cases}$

$= \begin{cases} 2x^2, & x \geq 0 \\ 0, & x < 0 \end{cases}$. Hence $y'(x) = \begin{cases} 4x, & x \geq 0 \\ 0, & x < 0 \end{cases}$

Note that we cannot find $\frac{dy}{dx}$ as the derivative does not exist at $x=0$

32. Given $t = \sqrt{6} - \sqrt{4} - \sqrt{3} + \sqrt{2} = \tan 7\frac{1}{2}$

$\therefore f(x) = \frac{2x}{1-x^2} \Rightarrow f(t) = \tan 15^\circ = (2 - \sqrt{3})$

$g(x) = \frac{3x-x^3}{1-3x^2} \Rightarrow g(t) = \tan\left(22\frac{1}{2}\right) = \sqrt{2} - 1$

Now $f(g(t)) = f(\sqrt{2} - 1) = \tan 2\left(22\frac{1}{2}\right) = \tan 45 = 1$

$\Rightarrow (f(g(t)))' = 0$

33. $f(x) = \cos x \cos 2x \cos 4x \cos 8x \cos 16x$

$= \cos A \cos 2A \cos 4A \dots \cos 2^n A = \frac{\sin 2^{n+1} x}{2^{n+1} \sin x}$

$= \frac{\sin 2^5 x}{2^5 \sin x} = \frac{\sin 32x}{32 \sin x}$

$f'(x) = \frac{32 \sin x \cos 32x - \cos x \sin 32x}{32 \sin^2 x}$

then $f'(\frac{\pi}{4}) = \frac{32 \sin \frac{\pi}{4} \cos 8\pi - 0}{32(\frac{1}{2})} = 2 \sin \frac{\pi}{4} = \cos \frac{\pi}{4}$

34. $f(x) = |x-8| = x-8$ as $x > 8$

Now $g(x) = f(f(x)) = f(x-8) = f(X) = X-8$

$g(x) = f(f(x)) = x-16 \quad \therefore g'(x) = 1$

35. Let $y^{y^{y^{\dots}}} = \log_e(x + \log_e(x + \dots)) = v$

$\therefore y^v = \log_e(x + v) = v \quad \Rightarrow y = v^x \text{ and } x = e^v - v$

$\Rightarrow \frac{dy}{dv} = v^x \left(\frac{d}{dv} \frac{1}{v} \log v \right) \text{ and } \frac{dx}{dv} = (e^v - 1)$

i.e. $\frac{dy}{dv} = v^x \left(\frac{1}{v} \cdot \frac{1}{v} - \frac{1}{v^2} \log v \right) = v^{x-2} (1 - \log v)$

and $\frac{dx}{dv} = (e^v - 1) \therefore \frac{dy}{dx} = \frac{v^{x-2} (1 - \log v)}{e^v - 1}$

at $x = e^2 - 2, y = \sqrt{2}, v^x = \sqrt{2}, e^v - v = e^2 - 2$

$\Rightarrow v = 2$ or 4 which is not valid

$\therefore v = 2, e^v - v = e^2 - 2 = x$ given so true

$\left(\frac{dy}{dx} \right)_{e^2-2, \sqrt{2}} = \left(\frac{dy}{dx} \right)_{v=2} = \frac{1 - \log 2}{2\sqrt{2}(e^2 - 1)}$

36. $y = \frac{\sin^3 x}{\sin x + \cos x} + \frac{\cos^3 x}{\cos x + \sin x} = \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$

$= \sin^2 x + \cos^2 x - \sin x \cos x = 1 - \frac{1}{2} \sin 2x$ then

$y'(x) = -\cos 2x$

37. By verification

38. $t = \sin^{-1}(2^s) \Rightarrow \sin t = 2^s$ differentiate w.r.t. 't' on both

sides $\cos t = 2^s \log 2 \cdot \frac{ds}{dt}$

$\Rightarrow \cos t = \sin t \log 2 \cdot \frac{ds}{dt} \Rightarrow \frac{ds}{dt} = \frac{\cot t}{\log 2}$

39. Express 'y' as $\frac{x}{x + \frac{x}{b+y}}$ and then differentiate

40. In the neighbourhood of $-\frac{\pi}{4}$, we have

$f(x) = (-x)^{-\sin x} = e^{-\sin x \log(-x)}$

$\Rightarrow f'(x) = e^{-\sin x \log(-x)} \left(-\cos x \cdot \log(-x) - \frac{\sin x}{x} \right)$

$\Rightarrow f'(x) = (-x)^{-\sin x} \left(-\cos x \cdot \log(-x) - \frac{\sin x}{x} \right)$

$\Rightarrow f'(-\frac{\pi}{4}) = (\frac{\pi}{4})^{(\frac{\pi}{4})} \left(\frac{-1}{\sqrt{2}} \log(\frac{\pi}{4}) + \frac{4}{\pi} \times \frac{-1}{\sqrt{2}} \right)$

$= (\frac{\pi}{4})^{(\frac{\pi}{4})} \left(\frac{\sqrt{2}}{2} \log(\frac{\pi}{4}) - \frac{2\sqrt{2}}{\pi} \right)$

41. $\tan^{-1}(x+1) - \tan^{-1}x + \tan^{-1}(x+2)$

$-\tan^{-1}x + 1 - \dots = \tan^{-1}x + n - \tan^{-1}x$

42. Put $x = \sin \alpha, y = \sin \beta$

43. we have $f(x) = 3e^{x^2}$, differentiating w.r.t. x, we get

$f'(x) = 6xe^{x^2}$

Now $f'(x) = 6xe^{x^2} - 2xf'(x) + \frac{1}{3}f(0) - f'(0)$

$= 6xe^{x^2} - 6xe^{x^2} + \frac{1}{3}(3) - 0 = 1$

$\Rightarrow f(0) = 3$ and $\Rightarrow f'(0) = 0$

44. $\frac{dy}{dx} = f' \left(\frac{2x+3}{3-2x} \right) \cdot \frac{d}{dx} \left(\frac{2x+3}{3-2x} \right) = \sin \left[\log \left(\frac{2x+3}{3-2x} \right) \right] \cdot \frac{12}{(3-2x)^2}$

45. $f(x) = \frac{\log(\log x)}{\log x^2}$ then differentiate w.r.t 'x' on both

sides and substitute $x=e$

46. $n=2$, verification

47) $\frac{d}{dy} \left(\frac{dx}{dy} \right) = \frac{d}{dy} \left(\frac{1}{\left(\frac{dy}{dx} \right)} \right)$

$$48. \frac{dy}{dx} = n \left(x + \sqrt{x^2 - a^2} \right)^{n-1} \cdot \left(1 + \frac{1}{2\sqrt{x^2 - a^2}} \cdot 2x \right)$$

$$= n \left(\frac{x + \sqrt{x^2 - a^2}}{x + \sqrt{x^2 - a^2}} \right)^n \left(\frac{x + \sqrt{x^2 - a^2}}{\sqrt{x^2 - a^2}} \right)$$

$$49. (1-x)^{-\frac{p}{q}} = 1 + p \left(\frac{x}{q} \right) + \frac{p(p+q)}{2!} \left(\frac{x}{q} \right)^2 + \dots$$

$$\Rightarrow \frac{d}{dx} (1-x)^{-\frac{p}{q}}$$

$$50. \frac{d}{dx} \left(\frac{\tan x - \frac{1}{\tan x}}{\tan x + \frac{1}{\tan x}} \right) = \frac{d}{dx} \left(\frac{-(1 - \tan^2 x)}{1 + \tan^2 x} \right)$$

$$= \frac{d}{dx} (-\cos 2x)$$

$$51. y = \tan^{-1} \left(\frac{1}{1+x(x-1)} \right) = \tan^{-1} \left(\frac{x-(x-1)}{1+x(x-1)} \right)$$

$$52. \frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2 - 1}}$$

$$53. \frac{d}{dx} \left(\coth \left(\frac{x}{2} \right) \right) = -\cosh^2 \left(\frac{x}{2} \right) \cdot \left(\frac{1}{2} \right) Z$$

$$54. \frac{dy}{d\theta} = n \sec^{n-1} \theta \cdot \sec \theta \tan \theta - n \cos^{n-1} \theta (-\sin \theta)$$

$$= n \tan \theta (\sec^n \theta + \cos^n \theta)$$

$$\frac{dx}{d\theta} = \sec \theta \tan \theta + \sin \theta$$

$$\left(\frac{dy}{dx} \right)^2 = \left[\frac{n \tan \theta (\sec^n \theta + \cos^n \theta)}{\tan \theta (\sec \theta + \cos \theta)} \right]^2$$

$$= \frac{n^2 (\sec^n \theta + \cos^n \theta)^2}{(\sec \theta + \cos \theta)^2}$$

$$= \frac{[n^2 (\sec^n \theta - \cos^n \theta)^2 + 4 \sec^n \theta \cos^n \theta]}{(\sec \theta - \cos \theta)^2 + 4 \sec \theta \cos \theta}$$

$$= \frac{n^2 (y^2 + 4)}{x^2 + 4}$$

55. Express 'y' as $\cot x - \cot(n+1)x$ and differentiate

$$56. 3f(x) - 2f\left(\frac{1}{x}\right) = x \text{ -----(1)}$$

$$\text{Put } x = \frac{1}{x} \text{ we get } 3f\left(\frac{1}{x}\right) - 2f(x) = \frac{1}{x} \text{ ---}$$

-- (2) Solve (1) and (2).

$$57. \log x = \tan^{-1} \left(\frac{y-x^2}{x^2} \right) y = x^2 + x^2 \tan(\log x)$$

58. Eliminate $f(\tan x)$

$$f(\cot x) = \frac{x}{b-a} - \frac{a\pi}{2(b^2 - a^2)}$$

$$59. \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

60. Let $g(x)$ be the inverse of $f(x)$

$$\Rightarrow g^1(f(x)) f^1(x) = 1$$

$$g^1(f(-1)) f^1(-1) = 1 \Rightarrow g^1(-2) = \frac{1}{14}$$

$$61. f(x) = \cos x \cos 2x \cos 2^2 x \cos 2^3 x \dots \cos 2^{n-1} x$$

$$\Rightarrow f(x) = \frac{\sin 2^n x}{2^n \sin x}$$

$$\Rightarrow f^1(x) = \frac{2^n \cos 2^n x \sin x - \sin 2^n x \cos x}{2^n \sin^2 x}$$

$$\Rightarrow f^1\left(\frac{\pi}{2}\right) = \frac{2^n \cos 2^{n-1} \pi}{2^n} = \cos 2^{n-1} \pi = (-1)^{2^{n-1}} = 1$$

$$62. g(x) = f^{-1}(x) \Rightarrow f(g(x)) = x$$

$$\Rightarrow f^1(g(x)) \cdot g^1(x) = 1$$

$$63. y = \{x^{11} + f(x)\}^{-2}$$

$$64. (fog)(x) = x \text{ for all } x$$

$$\Rightarrow f^1(g(x)) g^1(x) = 1 \text{ for all } x$$

$$\Rightarrow f^1(g(1)) g^1(1) = 1; \Rightarrow g^1(1) = \frac{1}{f^1(g(1))}$$

$$\text{But } (gof)(x) = x \Rightarrow g(f(0)) = 0$$

$$\Rightarrow g(1) = 0 \text{ and } f^1(0) = \frac{1}{2}$$

$$\text{Hence, } g^1(1) = \frac{1}{\frac{1}{2}} = 2$$

$$65. \frac{dy}{dx} = -\frac{f_x}{f_y}$$

$$66. f'(x) = \begin{vmatrix} 0 & 0 & 0 \\ \theta \sec x & \tan x & x \\ 1 & \tan x - \tan \theta & \theta \end{vmatrix}$$

$$+ \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec x \tan x & \sec^2 x & 1 \\ 1 & \tan x - \tan \theta & 0 \end{vmatrix} + \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec x & \tan x & x \\ 0 & \sec^2 x & 0 \end{vmatrix}$$

$$f'(0) = 0 + \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec \theta \tan \theta & \sec^2 \theta & 1 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec \theta & \tan \theta & \theta \\ 0 & \sec^2 \theta & 0 \end{vmatrix} = -1$$

$$67. \frac{du}{dx} = f'(x^2)(2x) = (\sin x^2)(2x)$$

$$\frac{du}{dx} = g'(x^3)(3x^2) = \cos(x^3)(3x^2)$$

$$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{2x \sin x^2}{3x^2 \cos x^3} = \frac{2 \sin x^2}{3x \cos x^3}$$

$$68. \frac{dy}{dx} = f'[f(f(f(x)))] \cdot f'[f(f(x))] \cdot f'(f(x)) \cdot f'(x)$$

$$\begin{aligned} \left(\frac{dy}{dx}\right)_{x=0} &= f'[f(f(f(0)))] \cdot f'[f(f(0))] \cdot f'[f(0)] \cdot f'(0) \\ &= f'[f(f(0))] \cdot f'[f(0)] \cdot f'(0) \cdot (2) \\ &= f'[f(0)] \cdot f'(0) \cdot (2) \cdot (2) \\ &= f'(0) \cdot (2) \cdot (2) \cdot (2) = (2) \cdot (2) \cdot (2) \cdot (2) = 16 \end{aligned}$$

$$69. \frac{dy}{dx} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dx}{dt}\right)} = \frac{\psi^1}{\phi^1}$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{\psi^1}{\phi^1} \right) = \frac{d}{dt} \left(\frac{\psi^1}{\phi^1} \right) \frac{dt}{dx} = \frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^2} \cdot \frac{1}{\phi^1} \\ &= \frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^3} \end{aligned}$$

$$70. \frac{dx}{ds} = 1, \frac{dy}{ds} = 2, \frac{d^2 x}{ds^2} = 0, \frac{d^2 y}{ds^2} = 0$$

$$u = x^2 + y^2 \Rightarrow \frac{du}{ds} = 2x \frac{dx}{ds} + 2y \frac{dy}{ds}$$

$$\begin{aligned} \frac{d^2 u}{ds^2} &= \frac{d}{ds} \left[\frac{du}{ds} \right] = \frac{d}{ds} \left[2x \frac{dx}{ds} + 2y \frac{dy}{ds} \right] \\ &= 2 \left(\frac{dx}{ds} \right)^2 + 2x \frac{d^2 x}{ds^2} + 2 \left(\frac{dy}{ds} \right)^2 + 2y \frac{d^2 y}{ds^2} \\ &= 2(1) + 0 + 2(4) + 0 = 10 \end{aligned}$$

$$71. t^8 = x - 1 \Rightarrow t^2 = (x - 1)^{1/4}$$

$$y = (t)^2 t^8 + 1 \Rightarrow y = (x - 1)^{5/4} + 1 \Rightarrow$$

$$\frac{dy}{dx} = \frac{5}{4}(x - 1)^{1/4}$$

$$\frac{d^2 y}{dx^2} = \frac{5}{16}(x - 1)^{-3/4} = \frac{5}{16(x - 1)^{3/4}} = \frac{5}{16(t^2)^3} = \frac{5}{16t^6}$$

$$72. \frac{d}{dx} \left[\cos^2 \left(\tan^{-1} \left(\sin \sin^{-1} \frac{1}{\sqrt{1+x^2}} \right) \right) \right]$$

$$\frac{d}{dx} \left[\cos^2 \left[\tan^{-1} \frac{1}{\sqrt{1+x^2}} \right] \right] = \frac{d}{dx} \left[\frac{1 + \cos 2 \left(\tan^{-1} \frac{1}{\sqrt{1+x^2}} \right)}{2} \right]$$

$$= \frac{d}{dx} \left[1 + \frac{1 - \frac{1}{1+x^2}}{1 + \frac{1}{1+x^2}} \right] = \frac{2x}{(2+x^2)^2}$$

$$73. x \sin y = \sin y \cos a + \cos y \sin a$$

$$\Rightarrow \sin y(x - \cos a) = \cos y \sin a$$

$$\frac{\sin y}{\cos y} = \frac{\sin a}{x - \cos a} \Rightarrow y = \tan^{-1} \left(\frac{\sin a}{x - \cos a} \right)$$

$$74. f(x) = \tan^{-1}(\cos x + n) - \tan^{-1}(\cos x)$$

$$f'(x) = \frac{1}{1+(n+\cos x)^2}(-\sin x) - \frac{1}{1+\cos^2 x}(-\sin x)$$

$$f'(0) = 0$$

$$75. f'(x) = \frac{-1}{1 + \left(\frac{x^x - x^{-x}}{2} \right)^2} \frac{d}{dx} \left(\frac{x^x - x^{-x}}{2} \right)$$

$$\begin{aligned} f'(1) &= \frac{-4}{(1+1)^2} \cdot \frac{1(1+\log 1) + 1(1+\log 1)}{2} = (-1) \left(\frac{2}{2} \right) \\ &= -1 \end{aligned}$$

$$76. \log y = x [\log(1 + 1/x)]$$

$$\frac{1}{y} y_1 = \frac{x^2}{x+1} \left(\frac{-1}{x^2} \right) + \log \left(1 + \frac{1}{x} \right) \Rightarrow y_1 = \frac{-y}{x+1} + y \log \left(1 + \frac{1}{x} \right)$$

$$y(2) = \frac{9}{4}; y_1(2) = \frac{9}{4} \left(\frac{-1}{3} + \log \frac{3}{2} \right)$$

$$y_2(2) = y_1(2) \left(\frac{-1}{3} + \log \frac{3}{2} \right) + y(2) \left(\frac{1}{9} - \frac{1}{6} \right)$$

$$y_2(2) = \frac{9}{4} \left(\frac{-1}{3} + \log \frac{3}{2} \right)^2 - \frac{1}{8}$$

77. $y = \frac{f(x)}{1 + \frac{g(x)}{1 + \frac{f(x)}{1 + \dots \infty}}} \Rightarrow \frac{dy}{dx} = \frac{(1+y)f'(x) - yg'(x)}{1+2y+g(x)-f(x)}$
78. $v+u=6\sqrt{uv} \Rightarrow v^2+u^2=34uv$ then differentiate
79. $y = \tan^{-1} \left(\frac{8 \sin x \cos x}{\cos^2 x - 7 \sin^2 x} \right)$
 $= \tan^{-1} \left(\frac{8 \tan x}{1-7 \tan^2 x} \right) = \tan^{-1} \left(\frac{7 \tan x + \tan x}{1-7 \tan x \cdot \tan x} \right)$
 $y = \tan^{-1} (7 \tan x) + x$
80. $y = \tan^{-1} \left(\frac{1-\log x^2}{1+\log x^2} \right) + \tan^{-1} \left(\frac{3+2 \log x}{1-3(2 \log x)} \right)$
 $= \tan^{-1}(1) - \tan^{-1}(\log x^2) + \tan^{-1}(3) + \tan^{-1}(3 \log x)$
 $= \tan^{-1}(1) + \tan^{-1}(3); \frac{d^2 y}{dx^2} = 0$
81. $y = \cot^{-1} \left[\frac{2+2 \cos x}{2 \sin x} \right] = \cot^{-1} \left[\frac{1+\cos x}{\sin x} \right] =$
 $\cot^{-1} \left[\cot \frac{x}{2} \right] = \frac{x}{2}; \quad \frac{dy}{dx} = \frac{1}{2}$
82. $y^1 = -\frac{x}{y} \Rightarrow y^{11} = \frac{-y+xy^1}{y^2} = -\frac{2}{y^3}$
 $-\frac{2}{y^3} + \frac{A}{y^3} = 0 \Rightarrow A = 2$
83. $p(x^2+1) = (p(x))^2 + 1 \Rightarrow p(x) = x,$
 $p^1(x) = 1 \Rightarrow p^1(0) = 1$
84. $e^{x^e} \cdot e \cdot x^{e-1} + x^{e^x} \left[e^x \cdot \frac{1}{x} + e^x \cdot \log x \right]$
 $+ e^{x^x} [x^x (1 + \log x)]$
85. Taking logarithms on both sides & differentiating
86. Taking logarithms on both sides & differentiating
87. $y = |\cot \theta| = -\cot \theta, \left(\theta = \frac{3\pi}{4} \right)$
88. Put $x^n = \sin \alpha, y^n = \sin \beta$
89. $y = -\cos x + \sin x$
90. $y = f(x^2) \Rightarrow \frac{dy}{dx} = f'(x^2) \cdot 2x$
 $= 2x\sqrt{2x^4-1}$

91. $f(x) = \frac{\pi}{2} - \sqrt{\frac{1+x}{2}} + x^x$
92. Differentiate
93. $[\because 0 \leq \cos^{-1} x \leq \pi \Rightarrow \text{in the neighbourhood of}]$
 $x = \frac{5\pi}{4}$ we have $0 < 2\pi - x < \pi]$
94. $\frac{dy}{dx} = f' \left(\frac{3x+4}{5x+6} \right) \frac{d}{dx} \left(\frac{3x+4}{5x+6} \right)$
95. $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$
96. Exponent of $x = \frac{l^2 - m^2 + m^2 - n^2 + n^2 - l^2}{(l-m)(m-n)(n-l)} = 0$
 $y = x^0 = 1$
97. $u(x) = 7.v(x) \Rightarrow u^1(x) = 7v^1(x) \Rightarrow p = 7$
 $\left(\frac{u(x)}{v(x)} \right)^1 = 0 \Rightarrow q = 0$
98. $f(x) = \cos x - \sin x, 0 < x \leq \frac{\pi}{4}$
 $\Rightarrow \sin x - \cos x, \frac{\pi}{4} < x < \frac{\pi}{2}$
99. $\frac{dx}{dt} = -\left(\frac{3}{t^4} + \frac{2}{t^3} \right) = -\left(\frac{3+2t}{t^4} \right)$
 $\frac{dy}{dt} = -\left(\frac{3}{t^3} + \frac{2}{t^2} \right) = -\left(\frac{3+2t}{t^3} \right); \quad \frac{dy}{dx} = t$
100. $f^1(x) = 0 + 0 + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix}$
101. Put $x = 1 \Rightarrow y = \frac{\pi}{2}$ and then differentiate.
102. $g^1(x) = 2(f(2f(x)+2)) \left(\frac{d}{dx} (f(2f(x)+2)) \right)$
 $= 2f(2f(x)+2) f^1(2f(x)+2) \cdot (2f^1(x))$
 $= g^1(0) = 2f(2f(0)+2) \cdot f^1(2f(0)+2) \cdot 2(f^1(0))$
 $= 4f(0) f^1(0)$
 $= 4(-1)(1) = -4$
103. $h^1(x) = 2f(x) f^1(x) + 2g(x) g^1(x)$
as $f^1(x) = g(x), \quad g^1(x) = -f(x)$
 $\therefore h^1(x) = 0 \quad \therefore h(x)$ is constant function.

104. we have, $f(x).f(y) = f(x) + f(y) + f(xy) - 2$

replace y by $\frac{1}{x}$

$$\Rightarrow f(x).f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + f(1) - 2$$

$$\Rightarrow f(x).f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

(since $f(1) = 2$ putting $x=y=1$)

$$\Rightarrow f(x) = x^n + 1 \Rightarrow f(2) = 2^2 + 1 \text{ (since } f(2)=5)$$

$$\Rightarrow n = 2 \therefore f(x) = x^2 + 1 \Rightarrow f(3) = 10$$

$$105. f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h}$$

$$= f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = f(x) \lim_{h \rightarrow 0} \frac{1 + (\sin 2h)g(h) - 1}{h}$$

$$= f(x) \lim_{h \rightarrow 0} \frac{\sin 2h}{h} \cdot \lim_{h \rightarrow 0} g(h) = 2f(x)g(0)$$



LEVEL - IV

1. A: $\frac{d}{dx} \left(\tan^{-1} x + \sin^{-1} \frac{x}{\sqrt{1+x^2}} \right) \text{ at } x=0$

B: $\frac{d}{dx} \left(\sec^{-1} \left(\frac{x^2 + \sqrt{x} + 1}{x^2 - \sqrt{x} - 1} \right) + \sin^{-1} \left(\frac{x^2 - \sqrt{x} - 1}{x^2 + \sqrt{x} + 1} \right) \right)$

C: $\frac{d}{dx} \left[\tan^{-1} \left(\frac{1 + \sin x}{\cos x} \right) \right]$

D: $\frac{d}{dx} \left(\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \right) \text{ at } x=1$

Arrangement of the above values in the increasing order of the magnitude

- 1) B, C, D, A 2) B, A, D, C
3) C, D, A, B 4) D, A, B, C

2. Observe the following lists.

List - I

List - II

A. The derivative of

$$\tan^{-1} \left(\frac{\sqrt{1+x^2} - 1}{x} \right)$$

1) $\frac{1}{1+x^2}$

w.r.t. $\tan^{-1} x$

B. $\frac{d}{dx} \left(x\sqrt{x^2-9} - 9\cos^{-1} \left(\frac{x}{3} \right) \right)$ 2) $\frac{2}{x}$

C. $\frac{d}{dx} \left(\tan^{-1} \left(\frac{2+3x}{3-2x} \right) \right)$ 3) $2\sqrt{x^2-9}$

D. The derivative of

$\sec^{-1} \left(\frac{1}{2x^2-1} \right)$ w.r.t. $\sqrt{1-x^2}$ is 4) $\frac{1}{2}$

5. $\sqrt{x^2-9}$

The correct match for list - I from list - II is

	A	B	C	D
1)	5	4	2	3
2)	4	5	3	2
3)	3	4	1	2
4)	4	3	1	2

3. Statement I : If $f(x) = \cos^{-1} \left(\frac{1 - (\log x)^2}{1 + (\log x)^2} \right)$

then $f'(e) = 2e^{-1}$

Statement II : If $f'(x) = 0 \forall x \in (a, b)$ then $f(x)$ is constant $\forall x \in [a, b]$

Which of the above statements is correct

- 1) Only I 2) Only II
3) Both I and II 4) Neither I nor II

4. Statement I : If $x = \sin \theta + \theta \cos \theta$,

$y = \cos \theta - \theta \sin \theta$, then $\left(\frac{dy}{dx} \right)_{\theta=\pi/2} = 1$

Statement II : If $x = \sec \theta - \cos \theta$,

$y = \sec^n \theta - \cos^n \theta$ then $\left(\frac{dy}{dx} \right)^2 = \frac{n^2(y^2 + 4)}{(x^2 + 4)}$

- 1) Only I is true 2) Only II is true
3) Both I and II are true 4) neither I nor II true

5. Statement I :

If $f(\theta) = \cos \theta_1 \cdot \cos \theta_2 \dots \cos \theta_n$

then the value of

$\tan \theta_1 + \tan \theta_2 + \dots + \tan \theta_n = \frac{-f'(\theta)}{f(\theta)}$

Statement II : Differential coefficient of the function

f(g(x)) w.r.t to the function g(x) is $f'(g(x))$

Which of the above statements is correct

- 1) I only 2) II only
3) Both I and II 4) Neither I nor II

6. Statement I : If $x = e^{y+e^{y+e^{y+\dots\infty}}}$ then

$$\frac{dy}{dx} = \frac{x}{1-x}$$

Statement II : If $y = |\cos x| + |\sin x|$ then the

value of $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is $\left(\frac{\sqrt{3}-1}{2}\right)$

Which of the above statements is correct

- 1) I only 2) II only
3) Both I and II 4) Neither I nor II

7. Observe the following

List - I

List - II

A. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$ then $\frac{dy}{dx}$ 1) $-\frac{1}{x^2}$

B. $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{2}$ then $\frac{dy}{dx}$ 2) $-\frac{x}{y}$

C. If $x^m y^n = (x+y)^{m+n}$ then $\frac{dy}{dx}$ 3) -1

D. If $\sin(x+y) = \log(x+y)$ 4) $\frac{y}{x}$

then $\frac{dy}{dx}$

The correct match is

	A	B	C	D
1)	2	1	3	4
2)	2	1	4	3
3)	1	2	3	4
4)	2	1	3	5

8. Assertion (A) : If

$$f(x) = \cos^2 x + \cos^2 \left(x + \frac{\pi}{3}\right)$$

$$- \cos x \cdot \cos \left(x + \frac{\pi}{3}\right), \text{ then } f'(x) = 0$$

Reason (R) : Derivative of constant function is zero

- 1) Both A & R are true, R is correct explanation of A
2) Both a & R are true, R is not correct explanation for A
3) A is true but R is false 4) A is false but R is true

9. a) $y = 2 \sin^{-1} \sqrt{\frac{x-\beta}{\alpha-\beta}}$ p) $\frac{dy}{dx} = \frac{4}{\sqrt{(\alpha-x)(x-\beta)}}$

b) $y = 4 \tan^{-1} \sqrt{\frac{x-\beta}{\alpha-x}}$ q) $\frac{dy}{dx} = \frac{1}{\sqrt{(\alpha-x)(x-\beta)}}$

c) $y = 4 \sin^{-1} \frac{2\sqrt{(\alpha-x)(x-\beta)}}{\alpha-\beta}$ r) $\frac{dy}{dx} = \frac{2}{\sqrt{(\alpha-x)(x-\beta)}}$

d) $y = 2 \cos^{-1} \sqrt{\frac{x-\beta}{\alpha-\beta}}$ s) $\frac{dy}{dx} = -\frac{1}{\sqrt{(\alpha-x)(x-\beta)}}$

a b c d a b c d

1) p q r s 2) q r p s

3) r s p q 4) s r q p

10. Observe the following lists.

List - I

List - II

A. $\frac{d}{dx}(\sin^{-1}(3x-4x^3))$ 1) $\frac{1}{2\sqrt{1-x^2}}$

B. $\frac{d}{dx}(\cot^{-1} \sqrt{\frac{1+\cos 3x}{1-\cos 3x}})$ 2) $\frac{-2}{1+x^2}$

C. $\frac{d}{dx} \left(\tan^{-1} \left(\frac{x}{1+\sqrt{1-x^2}} \right) \right)$ 3) $\frac{3}{\sqrt{1-x^2}}$

D. $\frac{d}{dx} \sin^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ 4) $\frac{2}{1+x^2}$

5) $\frac{3}{2}$

DIFFERENTIATION

The correct match for List - I from List - II is

A	B	C	D	A	B	C	D
1) 3	5	2	1	2) 4	3	5	1
3) 3	5	1	2	4) 4	2	5	3

11. Observe the following lists.

List - I

A. If $x = a(1 - \cos t)$;
 $y = a(t + \sin t)$ then

$$\frac{dy}{dx} \text{ at } x = \frac{\pi}{2}$$

$$B. \frac{d}{dx}(x^x) =$$

$$C. \frac{d}{dx} \left(\frac{x}{\sqrt{a^2 - x^2}} \right) =$$

$$D. \frac{d}{dx} \tan^{-1} \left(\frac{(3-x)\sqrt{x}}{1-3x} \right) =$$

List - II

$$1) x^x(1 + \log x)$$

$$2) \frac{a^2}{(a^2 - x^2)^{3/2}}$$

$$3) \frac{3}{2\sqrt{x}(1+x)}$$

$$4) 1$$

The correct match for List - I from List - II is

	A	B	C	D
1)	4	1	3	5
2)	3	1	2	4
3).	4	1	2	3
4)	4	1	2	5

$$12. A: \frac{d}{dx} \sin^2 \left[\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right]$$

$$B: \frac{d}{dx} \left(\frac{1 + \tan x}{1 - \tan x} \right) \text{ at } x = 0$$

C : For the curve at is

D :

Arrangement of the above values in the increasing order of the magnitude

- 1) A, B, C, D 2) C, A, D, B
3) C, D, B, A 4) A, D, B, C

JEE-MAIN-JR-MATHS VOL-IV

13. A : If $x = ct$, , at $t = 1$ is

$$B: x = 3 \cos \left(\frac{\pi}{2} \right) - \left(\frac{\pi}{2} \right), y = 3 \left(\frac{\pi}{2} \right)$$

Then is

C: If then

D : Derivative of $\log(\sec x)$ with respect to

$\tan x$ at is

Arrangement of the above values in the increasing order of the magnitude

- 1) C, D, A, B 2) C, A, D, B
3) A, B, D, C 4) B, D, A, C

14. A:

B:

C: at $x = 0$

D:

Arrangement of the above values in the increasing order of the magnitude

- 1) B, C, A, D 2) D, A, B, C
3) D, B, C, A 4) A, B, C, D

LEVEL-IV - KEY

- 1) 1 2) 4 3) 2 4) 2 5) 3 6) 2
7) 2 8) 1 9) 2 10) 1 11) 3 12) 2
13) 3 14) 2

LEVEL-IV - HINTS

1. A) Put

B)

- C)
- D) Put
2. A) Put
- B)
- C)
- D) Put
3. Differentiate and put $x = e$
4. St - I & St - II :
5. St - I : taking logarithms on both sides and then differentiate it
St - II : By the formula(42), it is correct.
6. St - I : By the formula (33)
St - II : Differentiate w.r.t. x
7. A)
- B)
- C) By taking logarithms and differentiate it
- D)
8. A :
- =
- =
- = 1 +

- then $f'(x) = 0$
- R : Derivative of constant function is zero
9. Apply inverse trigonometric formulae
10. A) Put B) formula
C) Put D) Put
11. A)
- B)
- C) Put D) Put
12. A) put , B)
- C)
- D)
13. A) B)
- C) D)
14. By the Formulae : A) formula 5, B) formula 13,
C) By the expansion of and find its derivative by the formula (2)
- D)